The Short- and Long-Run Consequences of Parental Violence Against Children

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Abstract

We study the short- and long-run consequences of violence against children perpetrated by their parents, combining detailed administrative data on police reports and healthcare use. We measure violence from police report data, which we combine with healthcare records and subsequent outcomes of the victims. To make progress on causal identification, we compare victims with matched controls, children recorded as victims at a later date, and siblings who are not recorded as victims. Victims and comparison children follow similar outcome paths before the event, and all three comparisons yield broadly similar short-run estimates afterward. Across the designs, we find that being victimized persistently increases mental health diagnoses after the event. In the long run, the victimized children have a reduced likelihood of graduating secondary education and an increased likelihood of being suspected and convicted of crimes themselves.

Keywords: Violence, Crime, Mental health, Human capital

JEL: J13, I14, K42, I21

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1 Introduction

What causes a human to behave violently? The cycle of violence, the idea that exposure to violence increases later violent behavior, is a central hypothesis across different fields (Widom, 1989; Caspi et al., 2002; Tremblay et al., 2018; Font and Kennedy, 2022). It links early-life environments to the development of aggression, crime, mental health, and human capital. Violence by parents against children is a particularly important form of exposure, as victimization by a caregiver is a severe adverse childhood experience with potentially long-lasting consequences. At the same time, it is strikingly common: in a 2022 Finnish survey, 9 percent of ninth-grade students reported being pushed, shoved, or shaken by a parent before age 14 (Mielityinen et al., 2023). Understanding its consequences is therefore important for explaining how violent and criminal behavior develops and why violence and crime persist within families.

In this paper we estimate the effects of parental violence against children. Violence rarely occurs in an otherwise happy and healthy household; it overlaps with parental mental illness, substance abuse, criminality, domestic violence, relationship instability, harsh parenting, and economic stress. These factors will influence later-life outcomes regardless of violence; hence, separating violence from the environment is a central issue. A second problem is that many datasets do not allow for a long follow-up of the victim children. Previous research has relied on survey data with retrospective measures of exposure and no measurements of pre-victimization outcomes. To make progress in understanding the consequences of violence in both the short and the long run, we combine multiple empirical designs and detailed administrative data.

Using detailed administrative data from Finland, we link police reports to suspected perpetrators and victims. While not all violence is observed and the data also include reports that do not ultimately lead to prosecution or conviction, they have important advantages. The data capture suspected offenses that never result in conviction and record their timing and the identities of the victim and suspected perpetrator without relying on retrospective survey reports. We link the reports to administrative data on healthcare use, demographics, education, labor-market outcomes, and convictions for the victims, suspected perpetrators, and other family members.

We define treatment as the first police report in which a child is recorded as the victim of a violent crime and a parent as the suspected perpetrator; event time is the earliest recorded date of the offense.[1] A recorded event may be followed by a police investigation, child-protection involvement, criminal sanctions, or family separation. The treatment therefore comprises the reported violence and the changes in the child's environment that follow it.

To make progress toward causal identification, we use multiple identification approaches. In addition to standard socioeconomic data, we observe detailed healthcare use. We match recorded victims to controls who have similar healthcare use and other characteristics before the first recorded event. We complement this design by comparing recorded victims with siblings who are not recorded as victims, again finding few pre-event differences. Our third design compares victims with children who are first recorded as victims later. None of these designs introduces a source of random variation in parental violence. While we proceed to use causal language in this paper, the assumptions required for a causal interpretation are strong. At the same time, we argue that this study makes progress toward understanding causal effects in a setting in which randomized controlled trials are not possible and credible instruments are hard to imagine.

Our three designs estimate different parameters. The matching and later-treated com-

[1] We also include sexual offenses, but most reports concern violence. We use "violence" below to refer to both violent and sexual offenses.

parisons include both the reported violence and subsequent changes that affect the family, such as child-protection involvement, criminal-justice consequences, and family separation. In contrast, the sibling comparison identifies the components of the event whose effects differ between the recorded victim and the sibling. A common family consequence (like the father leaving the household) is differenced out to the extent that it affects both children equally. Any difference in its effect on the victim and sibling remains in the estimate, so this does not provide a clean decomposition of the violence and other family-level events. The designs also differ in the assumptions under which they account for persistent determinants of victimization, so this can also mix up the interpretation of the results.

The first recorded event of parental violence coincides with a sharp and persistent increase in mental-health treatment. In the matched comparison, the gap in the probability of receiving any mental-health diagnosis widens by about 9.5 percentage points (compared with the control mean of 6.8%) between two years before the event and the following year. The increase remains visible throughout the follow-up period and also appears in antidepressant purchases, primary-care contacts, and specialized-care visits. The later-treated comparison shows a similar break at the first recorded event and no corresponding change in the preceding years.

We find similar absolute increases in the overall mental-health outcome for boys and girls, but the composition of recorded diagnoses differs: diagnoses of ADHD and conduct disorders increase more among boys, while mood and anxiety diagnoses increase more among girls. Criminal offending increases more among boys. Recorded mental-health diagnoses and treatment increase for children in all age groups. The increases are largest in relative terms and more persistent for younger victims, while older victims experience a sharp increase followed by a gradual decline.

The sibling comparison yields similar estimates. From two years before the event to the following year, the gap in any mental-health diagnosis between the recorded victim and a sibling who is not recorded as a victim widens by 8.8 percentage points. The siblings differ modestly before the event, but those differences are small relative to the change at the event. These outcomes measure diagnoses and healthcare contacts rather than mental health itself, and part of the initial increase may reflect detection or referral after a police report. The persistence of the increase, the rise in medication use, and similar changes among children already receiving mental-health care indicate that detection alone does not explain the pattern.

The differences extend into adulthood. Using the matching design, we find that victims are 20.6 percentage points less likely to complete secondary education by age 22, compared with a completion rate of 72.4 percent among matched controls. The sibling estimate is 14.3 percentage points, compared with 67.5 percent among siblings who are not recorded as victims. Annual wage earnings at age 25 are about €3,500 lower in the matching comparison and €3,235 lower in the sibling comparison.

The criminal-justice outcomes show the same pattern. By age 21, the probability of any conviction is 14.7 percentage points higher in the matched comparison and 9.8 percentage points higher in the sibling comparison. The corresponding increases in the probability of a violent-crime conviction are 7.5 and 5.1 percentage points. These estimates support the cycle-of-violence hypothesis: children recorded as victims of parental violence are substantially more likely to be convicted of crime themselves.

A large literature links childhood maltreatment to later crime, mental health, and economic outcomes (Widom, 1989; Currie and Spatz Widom, 2010; Currie and Tekin,

2012; Thornberry et al., 2010; Baldwin et al., 2023; Anderberg et al., 2025)[2]. Much of this evidence comes from surveys or comparisons made after maltreatment has occurred. The Finnish registers add the date of the first police-recorded event, outcomes measured before and after that date, and family links. These features allow us to show pre-event differences directly, estimate time-varying effects on recorded mental-health outcomes, and compare estimates from matched children, later-treated children, and siblings. The three comparisons do not remove every source of selection, but they make the assumptions and remaining differences visible. Overall, our exposure is probably on average more severe than the survey-reported exposures and we find broadly similar effects on adult outcomes as the closest methodological benchmark of Currie and Tekin (2012). We contribute to this literature with a different approach to identifying the effects, as well as an exploration of outcomes and heterogeneity absent from previous work.

We connect research on criminal victimization to research on the intergenerational persistence of crime. Recent work uses administrative data to study domestic violence and rape (Bhuller et al., 2024; Adams et al., 2024, 2026) and the intergenerational transmission of victimization and gender-based violence (Nix et al., 2026; Bhalotra et al., 2026).[3] A separate literature documents the persistence of criminal behavior from parents to children (Hjalmarsson and Lindquist, 2012, 2013). We show that violence by a parent against a child increases the child's own criminal behavior later in life, documenting a direct source for the intergenerational associations.

Finally, the paper connects research on human capital development with the economics of crime. Models of skill formation emphasize that childhood health and skills evolve jointly, so adverse conditions can affect later investments and adult outcomes (Cunha and Heckman, 2007; Currie, 2009; Almond et al., 2018). We add evidence from a severe event inside the family: the first police-recorded episode of parental violence. Linked registers allow us to follow the same children from mental-health treatment and school performance after the event to educational attainment, employment, earnings, and criminal convictions in early adulthood, and to compare these changes by age at first recorded victimization. Our results document how changes in health, education, work, and crime unfold after the recorded event—evidence relevant to economic theories in which human capital and labor-market opportunities affect criminal behavior (Becker, 1968). We also contribute to understanding of different types of 'criminal careers' (Moffitt, 1993; Nagin et al., 1995; Moffitt and Caspi, 2001) by showing evidence suggesting that exposure to violence leads to increases in both adolescent-limited and persistent offending.

The next two sections describe the Finnish criminal-justice, child-protection, healthcare, and education systems and the linked administrative records. We then define the matched, later-treated, and sibling comparisons and state the assumptions required for each. The results first trace healthcare use around the recorded event, then examine education, earnings, and criminal behavior in adulthood. The final section discusses what the estimates imply for the cycle of violence.

2 Institutional background

In this section, we describe the four features of the institutional setting that are most important for this paper: the criminal-justice system, child protection, healthcare, and the education system.

[2]Also related is the very large literature on other Adverse Childhood Experiences (ACEs) in psychology, medicine, and other fields (Baldwin et al., 2023; Kim and Royle, 2025). This literature is noted in reviews as needing more emphasis on identification of causal effects, which we aim to provide.

[3]The older economics literature is reviewed by Doyle and Aizer (2018)

2.1 Justice system

Criminal cases in Finland typically proceed in three stages. First, a suspected offence is reported to the police, who decide whether the threshold for a pre-trial investigation is met and, if so, carry out the investigation. The case file is then sent to the prosecutor, who decides whether to bring charges on the basis of the evidence collected during the investigation. Criminal cases are heard in district courts, with further appeals possible to a court of appeal and, subject to leave, to the Supreme Court.

For our purposes, the main distinction is between police reports and later stages of the process. A police report records suspected offending at a relatively low reporting threshold and therefore captures many incidents that never lead to prosecution or conviction. Prosecution and court outcomes require more evidence and are correspondingly narrower measures of violent offending. This is why police reports are particularly useful for measuring children's exposure to parental violence, while convictions are better interpreted as a selected measure of substantiated offending and criminal-justice response.

The legal definition of violence against children is broad. Corporal punishment is not a lawful form of discipline, and even relatively mild assaults against minors can constitute criminal offences. Since 2012, the Child Welfare Act has required public services (including early childhood education, schools, and health and social care) to notify the police when, in the course of their work, they have reason to suspect that a child has been subjected to a sexual offence. From 2015, this duty was extended to serious offences against life or health. In practice, these changes are likely to have lowered the threshold for child victimisation cases to enter police data.

2.2 Child protection and out-of-home care

Within Finnish child protection, not all involvement leads to a foster care placement. Many children remain at home and receive open-care support, while substitute care refers more narrowly to out-of-home care for children who are taken into care, placed urgently, or otherwise placed in substitute care. This distinction matters for our setting: foster care captures only a selected subset of child welfare cases.

When a child is placed outside the home, family care is the primary option. Institutional care is used when the child's needs cannot be met in family care or otherwise in the child's best interest. Placement decisions are case-specific and take account of the child's needs, close relationships, and practical continuity in everyday life such as schooling. For our period, the key point is simply that out-of-home care ends at adulthood, while child welfare aftercare may continue into early adulthood.

2.3 Healthcare

Finland has a universal public healthcare system that covers the entire population. Healthcare services, particularly for children and adolescents, are predominantly provided within the public sector. Privately provided healthcare is also available, but it is mainly accessed through employer-provided insurance and typically focuses on preventive care or treatment of mild conditions. As a result, private healthcare plays a limited role for the population studied in this paper.

Children receive frequent health checkups: annually at school and more often at child health clinics before school age. These checkups include physical- and mental-health screening. Although attendance is not mandatory, it is the default at schools, and child-protection services can be notified if parents repeatedly fail to take their children to the checkups.

Public healthcare involves modest out-of-pocket payments, which are capped at approximately 600 euros per year. A similar annual cap applies to prescription drug expenditures

for medications reimbursed by the Social Insurance Institute, which covers the vast majority of commonly prescribed drugs.

Healthcare records are an imperfect measure of underlying health status. We argue that, in our setting, the combination of broad coverage, frequent checkups, and low out-of-pocket costs limits differences in healthcare use arising from access to care.

2.4 Childcare and schooling in Finland

Finland provides publicly subsidized childcare, with no fees for low-income families and modest fees otherwise. Children typically move to preschool in the year they turn six, before beginning compulsory education at age seven. Childcare and preschool also provide regular contact with professionals who may detect suspected violence against children. Parents can also care for children under age three at home and receive the home-care allowance.

Compulsory education begins in the fall of the year when the child turns 7. For most cohorts in the long-run analysis, compulsory education ended after comprehensive school, typically around ages 15–16. For cohorts whose compulsory education ended in 2021 or later, the reform extended compulsory education to age 18 or completion of an upper-secondary qualification. Comprehensive schools are almost entirely publicly provided. There is no tuition, and meals as well as school materials are provided free of charge.

Students who continue to secondary education choose between an academic and a vocational track. Admission is primarily based on teacher-assigned grades at the end of ninth grade. In the academic track, students follow a standardized curriculum that includes mathematics, literature, languages, history, and natural sciences. Vocational education is organized into occupation-specific tracks, preparing students for professions such as car mechanic, plumber, or practical nurse. While the academic track is primarily intended to prepare students for university studies, both tracks allow progression to higher education. Secondary education typically lasts three years and is tuition-free, as is higher education.

3 Data

We combine population-wide Finnish administrative data on police reports, criminal convictions, healthcare use, labor-market outcomes, and demographics, linked across individuals and families using unique personal identifiers. The data were accessed and analyzed in secure facilities at Statistics Finland to ensure confidentiality and data protection. Information on data access is available from the author upon request.

3.1 Criminal-justice system data

We use two types of data from the criminal-justice system: police reports and convictions. Our main data source is the near-universe of police reports in Finland from 2006 to 2020. In principle, the system includes all reports filed with the police. Our data are limited to cases with a known suspected perpetrator under age 70. Reports with no known suspected perpetrator are excluded. For privacy reasons, the data also exclude very rare crimes, such as espionage, treason, and offenses under military law.

A report does not necessarily mean that the suspected crime is ever prosecuted or results in a conviction. The prosecutor decides whether to prosecute after the investigation, based on evidence gathered by the police.

The data include unique identifiers for the victim and the suspect, as well as some details of the crime and investigation. These include whether the police consider the case

as solved and whether the case was moved to the prosecutor. In line with previous studies, we do not condition on these outcomes in our main analysis.

All cumulative measures exclude convictions and reports that happened prior to the treatment [4] or during the same year.

3.2 Family violence against children

We identify family violence from police reports in which a child under age 18 is recorded as the victim and a parent as the suspected perpetrator.[5] We restrict the measure to violent and sexual offenses listed in Table A.2 and date the event using the earliest offense date recorded in the report rather than the report date.[6]

All violence against children is illegal during the study period. The reported cases are likely more severe than an average case of violence. This means that the victims in our sample are likely to have experienced more serious events than for instance in the previous studies relying on surveys. Indeed, the prevalence rates in our data are also lower than in surveys from Finland and abroad.

Compared with much of the previous research, which measures childhood exposure to violence using retrospective survey questions, our measure has several advantages. It is not subject to recall bias, which can be especially severe for events experienced by younger children. It also avoids attrition from survey sampling and response. Every case in our data has crossed a reporting threshold, whereas survey respondents must assess events that may have occurred many years earlier.

We mainly restrict the parental-violence measure to violence perpetrated by the victim's biological parents. The parent-child link is provided by Statistics Finland and is of high quality (Tilastokeskus, 2025b). These data also record whether the child lives with each biological parent. However, split custody is not reliably recorded here and in the cases in which a child does not live with father on the record, it can nonetheless be that the children live with the father for instance, every other week. We do not condition on this in our analysis, but use it as an outcome, so this should be kept in mind.

In parts of the analysis, we also consider crimes perpetrated by step-parents. We define step-parents as adults who are observed to be cohabiting with the biological parents. We obtain this linkage from the FOLK Cohabitation dataset (Tilastokeskus, 2025a). The dataset contains all cohabiting adults who have cohabited for at least 90 days and are not of the same sex. The dataset includes information on exact dates of moves as well as information on marriage between the partners. In practice, not all relationships are included in the dataset, nor are all cohabiting couples actually involved romantically. For our purposes, this matters little, since we are aiming to identify crimes within families.

3.3 Healthcare data

We measure healthcare use using nationwide administrative records from the public healthcare system. We use the HILMO register, which covers inpatient and specialized outpatient healthcare from 1996 to 2023, and the AvoHILMO register, which covers primary healthcare from 2011 onward. The records contain the dates of healthcare contacts and diagnosis codes, allowing us to measure both overall healthcare use and contacts associated with specific conditions. Our main mental health outcomes include indicators for receiving a mental health diagnosis and the number of healthcare contacts associated with mental

[4] Or quasi-treatment for the control group.

[5] We do not examine violence between siblings.

[6] The treatment includes the violence and the responses that can follow a report, although these responses need not occur on the date assigned to the first recorded event.

health diagnoses, as well as separate measures for particular diagnostic categories such as depression and anxiety.

As most of the healthcare, especially for children, is publicly provided, our healthcare measures are not selected based on provider choice. The out-of-pocket costs of a visit are low for adults and none for children, so there is also limited selection based on ability to pay. Still, individuals' healthcare use does not necessarily reflect their underlying health status.

A police report may lead to assessment and referral for treatment, so part of the immediate increase in recorded visits may reflect greater detection and treatment following the report rather than a change in underlying mental health. This issue is especially relevant for the youngest children, among whom mental-health diagnoses are rare and diagnostic manuals are often not designed for these cases.

We complement the healthcare records with administrative data on reimbursed prescription-drug purchases. These data cover prescriptions originating from both public and private healthcare, provided that the prescription is filled. We observe the timing of purchases, expenditures, and ATC codes, which allow us to identify purchases of antidepressants and other classes of mental-health medication. In the empirical analysis, we use both indicators for any purchase within a given drug class and measures of the number of purchases.

In addition to the victims, we also link the healthcare data to the parents and siblings of the victims to understand changes in the entire family around the family-violence event.

3.4 Demographic and Labor Market data

We use information on individual earnings, labor-market status, family, place of residence, and demographics from Statistics Finland's FOLK BASIC and FOLK INCOME datasets (Tilastokeskus, 2025c,d). These data provide a detailed view of economic activity, social-security use, and family composition. Most information is available annually for the entire population from 1987 to 2023.

3.5 Education data

The Finnish school system does not use frequent standardized testing. We observe grades at the end of compulsory education (ages 15–16) from the register of applicants to secondary education. The grades are assigned by teachers and are based on performance during the entire school year rather than a single test. They are relatively high-stakes because they are used in applications to secondary education. Although the grades are not standardized, they are the best available measure of educational performance. We use both the raw grades and a measure standardized within school $\times$ year to remove grade inflation.

We also observe whether an individual graduated from secondary or higher education using the information on the highest obtained degree from the FOLK BASIC dataset.

3.6 Sample selection

A child is treated in the year of the earliest recorded offense date in the first police report that identifies the child as a victim of parental violence. We restrict the sample to events observed between 2006 and 2020. From 2009 onward, we retain reports in which the complainant is recorded as the victim; this field is unavailable before 2009.

We further restrict the sample to children first recorded as victims between ages 2 and 17, so that we observe at least some pre-event outcomes. Children may experience subsequent parental violence after this first event, and they may also be victimized outside the family, although such cases are rare. We drop children for whom either biological

parent is missing from the registers in the treatment year, for example because the parent has died or emigrated. These restrictions yield approximately 18,000 treated children. The exact-matching sample requires three eligible controls per victim. The exact-plus-propensity-score sample, which supplies the main adult estimates, retains one control per victim and excludes victims without an eligible control. This procedure matches 92.2 percent of otherwise eligible victims. The sibling design includes victims with at least one sibling who is never recorded as a victim of parental violence.

Unless mentioned, we do not limit the analysis to a balanced sample. For long-run outcomes measured at a fixed age, the available sample disproportionately consists of children victimized at older ages and in earlier calendar years, since they must have reached the relevant age by the end of the observation period.

Observations cease after death or emigration, but we do not restrict the baseline sample to individuals who remain alive and resident throughout the follow-up.

3.7 Outcomes

We study outcomes in four domains: health, education, labor-market activity, and criminal behavior.

Our main earnings outcome is annual wage earnings, expressed in 2020 euros.

We observe graduation from secondary and higher education, as well as ninth-grade GPA. We use graduation from secondary education by age 22 and from higher education by age 25 as our main outcomes. The age-25 cutoff trades off follow-up length against sample size. Many university students graduate after age 25, so this measure does not capture lifetime graduation rates.

From the healthcare data, we measure several indicators of recorded mental-health diagnoses and treatment. We also use diagnoses that are not plausibly affected by the event as placebo outcomes and measure antidepressant purchases in the drug-reimbursement data. The outcomes include indicators for any diagnosis and the annual number of visits associated with a particular diagnosis. For prescriptions, we use an indicator for any purchase and the number of purchases within a particular drug class.

From the criminal-justice data, we use both police reports and convictions as outcomes. Convictions are a narrower measure of criminal offending because a case must progress further through the criminal-justice system.

Using these data, we measure the treatment and construct outcomes for recorded victims in adolescence and adulthood. We use both convictions and police reports. The main outcomes are: 1) Cumulative probability of any conviction by a particular age. Includes all crimes in our data.
2) Cumulative number of convictions by a particular age.
3) Cumulative probability of any police report by a particular age.
4) Similar measures for violent crime.

We examine violent crime separately because the psychological and biological literature identifies violent offending as a behavior that may be transmitted across generations.

The long-run cumulative crime outcomes and ninth-grade GPA are measured only after the first recorded event. When using these variables as outcomes, we treat them as missing before the event year.

3.8 Classifying criminal careers

A sizable literature in psychology and criminology (Moffitt, 1993; Moffitt et al., 2002; Nagin and Land, 1993) distinguishes among different patterns of criminal offending. In

particular, it proposes that many people offend during adolescence but do not continue along this path. This group is commonly termed "adolescent-limited offenders." Another group, "life-course-persistent offenders," displays signs of behavioral problems at young ages and continues offending through adolescence and into adulthood.

Economists sometimes recast the LCP/adolescence-limited distinction as a dynamic interaction between persistent individual heterogeneity and state dependence, where early criminal behavior can alter subsequent schooling and crime choices (Mancino et al., 2016; Beatton et al., 2018; Fu et al., 2022; Bell et al., 2022). A related strand emphasizes age of onset and the schooling–crime feedback, finding that earlier delinquency or arrest has particularly large effects on educational trajectories (Ward et al., 2021).

We use sequence analysis to explore these criminal careers and analyze whether victims are more likely to follow a particular offending trajectory. Sequence analysis can classify individuals based on the similarity of their trajectories and is frequently used in sociology and demography (Abbott, 1995; Aisenbrey and Fasang, 2010).

For each individual, we construct a yearly sequence from ages 15 to 25. In each year, the individual is classified as having committed no recorded crime or as having been suspected or convicted of at least one crime. We then calculate distances between these sequences and cluster individuals with similar trajectories. The precise construction of the sequences, distance measure, clustering procedure, and choice of the number of clusters are described in Section A.1.1.

Fig. A.3 displays the rates of crime in these different clusters. We denote cluster 1 as "No recorded offending," cluster 2 as "Infrequent offending," cluster 3 as "Adolescent-limited offending," and cluster 4 as "Persistent offending." Individuals in the infrequent-offending cluster have occasional recorded offending but no clear concentration at a particular age. Individuals in the adolescent-limited-offending cluster exhibit substantially more offending during adolescence and early adulthood, followed by a pronounced decline. Individuals in the persistent-offending cluster have high rates of recorded offending throughout the observation period.

These labels are descriptive, and sequence analysis is a method for measuring and classifying criminal careers rather than for causal inference. We subsequently use cluster membership as an outcome to examine whether parental violence changes the distribution of individuals across these criminal careers. These clusters are not constructed from the same measures as the categories in the foundational criminology literature (Moffitt, 1993), but they identify clearly different paths in our adolescent sample.

4 Methods

Parental violence occurs in families that differ from other families in many observed and unobserved respects. We use three empirical designs with different comparison groups and identifying assumptions:

***Matching*:** We use pre-event characteristics to construct a control group. The design estimates short-run changes and adult differences but remains vulnerable to unobserved differences between victims and controls.

***Comparison to later treated*:** We compare recorded victims with individuals whose first recorded victimization occurs later. Conditional on the included covariates, identification requires that the timing of first recorded victimization is unrelated to counterfactual outcome changes.

***Comparison to siblings not recorded as victims*:** We compare recorded victims with siblings who share a mother but are not recorded as victims of parental violence.

Mother fixed effects remove persistent characteristics common to siblings but not differences between siblings or differential responses to family events.

Agreement across the designs is informative only after accounting for the different effects that they estimate.

Parameters of interest

The event observed in the police data is the first report that records a child as the victim of parental violence, dated by the earliest recorded offense date. This event may combine several changes that could affect the child: the violence itself, reactions from other family members, police intervention, a child-protection investigation or other child-protection services,[7] healthcare contact, criminal sanctions, and family separation.

Some subsequent events may affect every child in the family. Parental separation or a parent's conviction, for example, can affect both the recorded victim and a sibling, although not necessarily to the same degree. Other components, including the violent act, urgent mental-health treatment, or an individual child-protection placement, may affect only the recorded victim.

Under conditional independence, the matching estimate combines the reported violence with the differences in subsequent family and institutional responses between victims and matched controls.[8]

The sibling estimate captures the components of the event whose effects differ between the recorded victim and the sibling. These components include the violence directed at the victim and any additional response directed toward that child. Common family consequences, such as family separation, child-protection involvement, or a parent's conviction, are differenced out only to the extent that they affect both siblings equally. Differences in the effects of these consequences remain in the estimate. This interpretation also requires that victimization is measured without error.

Finally, the later-treated control group has not yet experienced the recorded event at the time of comparison, so the treatment includes the same combination of violence and subsequent responses as in the matching design. By restricting the comparison group to children who are eventually recorded as victims, this design accounts for permanent factors associated with selection into recorded victimization. Conditional on the included covariates, identification requires that the timing of first recorded victimization is unrelated to counterfactual outcome changes. The design offers another estimate of the same parameter as the matching design in the short run.

4.1 Matching

In our main analysis, we estimate the consequences of the first police-recorded event using matched controls. We construct two matched samples. The exact match retains three controls for each victim and is used to show outcome paths before and after the event. The main adult estimates use exact matching followed by propensity-score matching and retain one control per victim.

Both samples first match exactly on event year (two-year bins); the child's age in two-year categories and gender; each parent's age in five-year categories; the mother's and father's education; whether the father was born in Finland and worked in a blue-collar occupation; whether the child lived with the mother and father; whether the child

[7] In addition to foster care, child protection provides services that do not involve removing the child from the family.

[8] Family separation and other events can also occur among controls. The estimate includes the difference in their incidence following the recorded event.

had a mental-health contact in the year before the event; whether the mother purchased mental-health medication and whether the father had a criminal conviction during the three preceding years; and whether the mother had low employment during the five preceding years. The exact-matching specification selects three controls within these cells. It does not use the full pre-event path of the outcomes, so the event studies can assess whether victims and controls followed similar trends before the recorded event.

The exact-plus-propensity-score specification uses the same exact cells and, within each cell, selects the control with the closest propensity score. Following Adams et al. (2026), we estimate the propensity score with a linear probability model. It includes the child's pre-event mental-health diagnoses and contacts; parental mental-health medication, addiction-related healthcare, criminal convictions, victimization, employment, marital status, co-residence, earnings, and social assistance; an indicator for a parent's prior violent conviction; and calendar-year terms. Variables without an explicit history are measured in the year before the event. Three-year histories cover event years -3 through -1, two-year histories cover years -2 through -1, and five-year histories cover years -5 through -1. Table A.3 lists the variables as they appear in the matching code. This specification matches 92.2 percent of otherwise eligible victims. We use it for the main adult estimates in Tables A.5, 3 and 4 and also report it alongside exact matching in the childhood event studies.

Each victim and the associated control or controls form a matching cell. The regressions include matching-cell fixed effects. Because the comparison is defined within cells, the estimates do not use already-treated children as controls in a two-way fixed-effects staggered-adoption regression (Goodman-Bacon, 2021).

Propensity-score matching uses more of the pre-event information but makes the comparison of pre-event outcome paths partly mechanical. We therefore report childhood event studies for both the exact match and the exact-plus-propensity-score match. The adult estimates use the latter because the adult outcomes are not observed before treatment.

Following Adams et al. (2026), we estimate time-varying outcomes using

$$Y_{it} = \sum_{j=-7}^{10} \beta_j \cdot Victim \cdot 1\{Time = j\} + \sum_{j=-7}^{10} \alpha_{cj} + \gamma_t + \epsilon_{it} \tag{1}$$

where α_{cj} denotes matching-cell-by-event-time fixed effects and γ_t contains calendar-year and child-age fixed effects; regressions for parental outcomes also include the parent's age. The coefficients β_j compare victims and controls at each event time. Unlike a standard difference-in-differences event study, the specification does not normalize the difference in one pre-event year to zero. The pre-event coefficients therefore show both level differences and changes in those differences.

For adult outcomes observed only after the event, we cannot assess pre-event outcome differences or trends. Identification therefore depends more directly on the matching variables and the conditional-independence assumption.

The matching design nevertheless permits estimates for adult education, earnings, and criminal convictions. For an outcome measured at a fixed age, we estimate

$$Y_{it} = \beta Victim_i + \alpha_c + \gamma_t + X_{it} + \epsilon_{it} \tag{2}$$

where α_c denotes matching-cell fixed effects, γ_t denotes calendar-year fixed effects, and X_{it} contains additional pre-event controls in some specifications. The coefficient β is the conditional difference between victims and controls.

Table 1 reports means, standardized differences, and tests of equality for the exact-plus-propensity-score sample. Several pre-event characteristics remain imbalanced. The

standardized difference is 0.150 for a maternal conviction during the preceding three years and −0.110 for paternal employment; the standardized differences in maternal and paternal earnings are −0.070 and −0.089. Because these characteristics predict the education and crime outcomes studied below, we report specifications that add pre-event parental earnings, employment, occupation, criminal history, victimization, healthcare use, and the child's pre-event mental-health measures. Although some covariates remain imbalanced after matching, adding these characteristics directly as controls leaves the adult estimates very similar (Fig. A.14). We also compare estimates across age-and-gender matching, exact matching, exact-plus-propensity-score matching, and exact-plus-propensity-score matching with these additional controls in the same figure.

	Mean			
Characteristic	Matched controls	Victims	Standardized difference	p-value
Panel A: Parents and household				
Mother's age (years)	36.4	36.2	−0.022	< 0.001
Father's age (years)	39.5	39.7	0.017	< 0.001
Mother's wage earnings (euros)	15,922	14,695	−0.070	< 0.001
Father's wage earnings (euros)	27,490	25,286	−0.089	< 0.001
Mother employed	0.595	0.576	−0.038	< 0.001
Father employed	0.746	0.697	−0.110	< 0.001
Mother lives in an urban area	0.858	0.870	0.036	< 0.001
Mother in a blue-collar occupation	0.428	0.400	−0.057	< 0.001
Age of mother's youngest child (years)	5.50	5.15	−0.080	< 0.001
Mother convicted in previous three years	0.017	0.042	0.150	< 0.001
Child lives with mother	0.932	0.932	0.000	—
Child lives with father	0.624	0.624	0.000	—
Panel B: Child health and healthcare use				
Specialized-care mental-health visits	0.534	0.692	0.043	< 0.001
Annual healthcare costs (euros)	82.4	72.5	−0.011	0.305
Any central-nervous-system prescription	0.034	0.036	0.013	0.088
Any specialized-care anxiety diagnosis	0.011	0.011	−0.003	0.733
Any specialized-care depression diagnosis	0.008	0.011	0.022	0.018
Any developmental-disorder diagnosis (F8)	0.037	0.033	−0.018	0.026
Any behavioral or emotional disorder diagnosis (F9)	0.037	0.045	0.043	< 0.001
Primary-care mental-health visits	0.100	0.102	0.002	0.874
Any primary-care mental-health visit	0.041	0.041	0.001	0.907
Any addiction-related healthcare visit	0.001	0.002	0.009	0.396
Observations	17,592	17,592		

Notes: The table compares children recorded as victims of family violence with their matched controls. Characteristics are measured in the year before the event, except the mother's conviction indicator, which covers the previous three years. Entries are shares unless the row specifies euros, years, visits, costs, or reports. The standardized difference is the difference in group means divided by the pooled standard deviation. The p-value tests equality of the two group means. HILMO covers inpatient and specialized outpatient care and AvoHILMO primary-care visits.

Table 1: Summary statistics of the main matching controls and the treated.

With heterogeneous effects, the matching estimate need not equal the effect of a specific intervention that prevents parental violence or changes the response after a report.

Matching cannot establish that the included variables capture all selection into recorded victimization. Parents who are suspected of these offenses may differ from other parents in unobserved ways that also affect their children's outcomes.

Events occurring near the first police-recorded episode may also contribute to the estimates. Parental employment does not deteriorate before the event, but maternal mental-health treatment and the mother's own victimization change near the event year.

The long-run estimates require that, conditional on the matching variables and fixed effects, the remaining differences between victims and controls are unrelated to their untreated adult outcomes. We assess the sensitivity of the estimates to alternative matching variables and to the additional pre-event controls described above. Table A.3 documents the matching specifications, and Fig. A.14 compares the resulting adult estimates.

We next present the later-treated and sibling comparisons.

4.2 Comparison to later treated

Our second design compares children who are first recorded as victims in different years. Let G_i denote the year in which child i is first recorded as a victim of parental violence. We treat the child as treated from year G_i onward, whether or not another police report is recorded. For each two-period comparison, the controls are children who have not yet been recorded as victims in either year. A child stops serving as a control when the child's first recorded victimization occurs. Children who are never recorded as victims are not included in this design.

We estimate group-by-time average treatment effects using the method of Callaway and Sant'Anna (2021). For each treatment cohort g and calendar year t, the estimator compares the change in the outcome between years $g-1$ and t with the corresponding change among later-treated children. We condition on the child's gender and birth year. We then aggregate these estimates by event time, $j = t - g$. The event study covers event times -5 through 5, with event time -1 normalized to zero.

We do not require children to be observed at every event time from -5 through 5. Instead, we balance the panel separately for each two-period comparison, retaining children observed in both years used in that comparison. The children and treatment cohorts contributing to the estimates can therefore differ across event times.

Identification requires that, conditional on gender and birth year, outcomes in the absence of the first recorded victimization would have changed similarly for children victimized earlier and later. It also requires comparable later-treated children within these characteristics. Using only later-treated children means that we do not compare victims with children who are never recorded as victims. It does not remove permanent or time-varying differences between children victimized earlier and later. The main specification also assumes that outcomes do not change in anticipation of the first recorded victimization. The pre-treatment estimates show whether outcome changes are similar before victimization, but they cannot prove that the trends would have remained similar afterwards.

Because violence may precede the first event recorded in the police data, we also allow outcomes to change one year before the first recorded victimization. In this specification, children stop serving as controls one year before their own first recorded victimization. Event time -2 is the reference period. Event time -1 is allowed to be affected and is not used as a pre-treatment test.

4.3 Sibling comparison

The matching design may leave unobserved differences between families, while the later-treated comparison cannot estimate adult outcomes because every child in its sample is eventually recorded as a victim. The sibling design compares recorded victims with siblings who are not recorded as victims and therefore permits both childhood and adult estimates. Mother fixed effects remove persistent family characteristics shared by siblings and part of their shared genetic background. The design does not remove differences between siblings and estimates a different parameter from the matching design, as described in Section 4.

The main specification uses every sibling who shares the victim's mother and is never recorded as a victim of parental violence. Victims without such a sibling are excluded.[9] If a family contains more than one recorded event, event time is defined by the first event in the family.

[9] We also exclude them from the balance checks so that those checks use the estimation sample.

The event study equation analogous to Eq. (1) is

$$Y_{it} = \sum_{j=-7}^{10} \beta_j \cdot Victim \cdot 1\{Time = j\} + \sum_{j=-7}^{10} m_{cj} + \gamma_t + \epsilon_{it} \tag{3}$$

where m_{cj} is a family-by-relative-time fixed effect, γ_t denotes calendar-year and age fixed effects, and β_j is the difference between recorded victims and siblings who are not recorded as victims. The control siblings are on average born earlier than the treated siblings, which the age fixed effects take into account. The specification does not eliminate spillovers within the family. A consequence that affects both siblings equally is differenced out, while any difference in its effect on the recorded victim and the sibling remains in β_j. The main specification includes mother fixed effects; as a robustness check, we restrict the comparison to full siblings. As in the matched sample, the event study also shows whether the siblings differ in outcome levels before the event.

For the long-run outcomes we use the version

$$Y_{it} = \beta Victim_i + m_c + \gamma_t + X_i + \epsilon_{it} \tag{4}$$

where m_c denotes mother fixed effects, γ_t denotes calendar-year fixed effects, and X_i contains pre-event individual characteristics in some specifications.

Sibling comparisons require additional assumptions about which child is recorded as a victim, treatment measurement, and spillovers within the family.

First, which sibling is recorded as a victim might not be random (Frisell et al., 2012). School performance, disability, or behavioral problems may affect both the risk of parental violence and later outcomes. Such differences would bias the estimates in the direction of the observed education and crime differences. A twin comparison would remove age-specific family circumstances, but it would not remove these child-specific differences, and our sample does not support a separate twin analysis.

We observe siblings before the recorded event and can therefore measure some of these differences. Recorded victims have higher pre-event rates of ADHD and conduct-disorder diagnoses. The event-study specification permits these outcome levels to differ, but its interpretation requires that the difference would not have changed sharply in the event year without the recorded event. For the adult outcomes, we add the pre-event health measures as controls; the estimates change little. Unobserved child-specific differences may nevertheless remain.

	(1) Control mean	(2) Victim mean
Birth year	2002.54	2006.00
Age at measurement	9.65	6.26
Single parent	0.27	0.32
Hosp. visits	1.73	1.85
Any MH diagnosis	0.06	0.07
Any anx. diag.	0.01	0.01
Any Depr. diag.	0.01	0.00
Antidepr. presc.	0.05	0.01
N diff prescr.	1.22	1.21
Mother wage earnings	19384.79	19439.96
Mother employed	0.53	0.53
Mother secondary	0.54	0.53
Mother HE	0.15	0.18
Observations	19434	18954

Table 2: Summary statistics of recorded victims and comparison siblings who are not recorded as victims.

Second, siblings who are not recorded as victims may nevertheless experience parental violence. Within-family estimates can be especially sensitive to such treatment misclassification (e.g. Griliches, 1979). Classical measurement error would attenuate the estimate toward zero (Ashenfelter and Krueger, 1994; Bound and Solon, 1999).

Third, families contributing within-family variation may differ from other families, and the remaining within-family variation may not represent differences in parental violence across families. Miller et al. (2023) describe this as selection into identification. Engzell and Hällsten (2025) discuss the issue for exposures that primarily vary between families.

Matching estimates are similar for victims with and without an eligible comparison sibling, which indicates that the restriction to families with within-family variation does not explain the difference between the matching and sibling estimates.

Finally, events following the report may affect the comparison sibling (Sjölander et al., 2016). The sibling estimate removes a spillover only when it affects the recorded victim and the sibling equally. If family separation, child-protection involvement, a parent's conviction, or another response affects the two children differently, that difference remains part of the estimate. We therefore interpret the sibling results as the combined effect of all components of the recorded event that differ between siblings, not as the effect of the violent act alone (Section 4).

5 Results

5.1 Matching

5.1.1 Effects in childhood

Victimization has large and persistent effects on recorded mental-health diagnoses and healthcare use. Figure 1 presents the main results for two outcomes: an indicator for any mental-health visit and the number of mental-health visits. The figure first displays the raw means and then the coefficients from Eq. (1). It presents event studies for two matched control groups, one based only on the exact-matching variables and one that also uses the propensity score. We prefer the propensity-score control group for the long-run outcomes, but the exact-matching control group is more useful for assessing pre-event paths because it does not match on these paths.[10]

Reassuringly, victims and controls follow similar pre-event paths, both in the raw means and in the regression estimates. The control group shows no corresponding break around the first recorded event.

The post-event estimates show a large and persistent increase in recorded mental-health diagnoses and treatment after victimization. The probability of having any mental-health diagnosis increases by around 5 percentage points in the year of the first recorded event and by a further 5 percentage points in the following year.[11]

We also estimate effects on more granular healthcare-use outcomes. Figures A.4 and A.5 display a similar set of graphs for antidepressant prescriptions, primary-care visits with mental-health diagnoses, and specialized-care visits with recorded diagnoses for neurotic and anxiety disorders (ICD-10 F4), mood disorders and depression (ICD-10 F3), and behavioral and emotional disorders (ICD-10 F9). All these outcomes show similar pre-event paths followed by a sharp and persistent increase. Antidepressant purchases and primary-care visits come from different datasets and represent different types of healthcare use.

[10] Propensity-score matching does not match directly on the pre-event paths either, but it uses this information in the matching.

[11] Because these are calendar years, individuals are first recorded as victims at different times during the year, which partly explains this pattern.

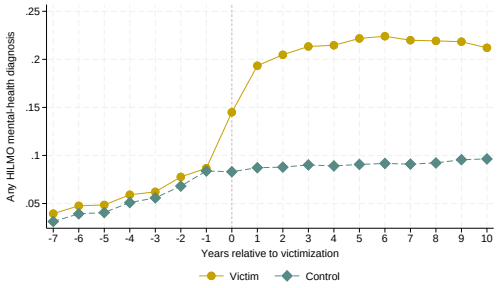
The similar patterns across these measures are consistent with an increase that extends beyond a single source of recorded diagnoses.

Figure A.6 shows estimates for healthcare use more generally. We measure the number of hospital and specialized-care visits, the number of primary-care visits, and hospital or specialized-care visits with a physical-injury diagnosis (ICD-10 S). Victims and controls differ somewhat in the pre-event levels of these outcomes, but their paths are similar. All outcomes increase sharply at the first recorded event.

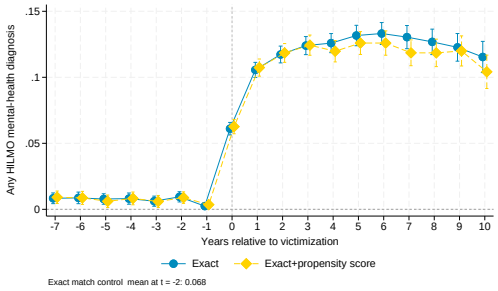
Mental-health diagnoses are uncommon among young children and do not directly measure latent mental health. Detection and referral likely explain part of the immediate increase, but several patterns make a purely detection-based explanation unlikely. Mental-health visits remain elevated for many years, and antidepressant purchases also increase persistently. We also find similar increases among children who had already received mental-health treatment before the event. Together, these patterns indicate that the post-event increase is not limited to an initial diagnostic response.

As we describe in Section 2, the welfare-state setting with universal health insurance also limits the scope for these issues. Because healthcare is free of charge for children, barriers to accessing care are smaller than in many other settings. The public healthcare system also provides health checkups with nearly universal attendance. These features do not rule out changes in healthcare access following the event, but they make access less likely to explain the full increase.

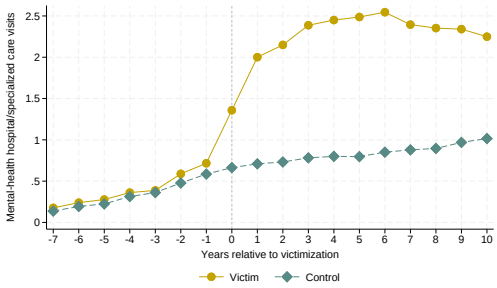
The similarity of pre-event healthcare-use outcomes between victims and controls supports using the same matching procedure to estimate adult outcomes.



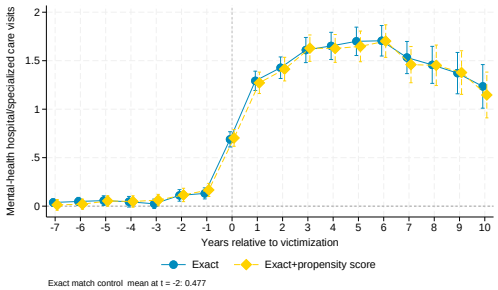
(a) Raw means: Any mental health diagnosis



(b) Estimates: Any mental health diagnosis



(c) Raw means: N of specialist visits with MH diagnosis.



(d) Estimates: N of specialist visits with MH diagnosis.

Figure 1: Raw means and matched event-study estimates using Eq. (1). The matching variables are listed in Table A.3. Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event control mean.

5.1.2 Effects in adulthood

We next present the estimates of the long-run effects. As no pre-trends for these outcomes are available, these results rely on stronger assumptions than the short-run results and should be read with these caveats in mind.

Victims have worse outcomes across all adult domains examined. Table 3 presents the economic and educational outcomes. The largest differences appear in extensive-margin outcomes such as secondary-school completion. The probability of completing secondary education by age 22 declines by 20.6 percentage points, and ninth-grade GPA declines by 0.372 grade points.

We next examine evidence for the "cycle of violence"—that is, whether exposure to parental violence increases the child's own subsequent criminal offending. The measures of criminal offending also increase (Table 4). Both police-reported crime and criminal convictions rise. The probability of being convicted of at least one crime by age 21 increases by 14.7 percentage points, to more than twice the control mean.

Violent crime also increases. The probability of being convicted of at least one violent crime by age 21 rises by 7.5 percentage points. Outcomes that combine the intensive and extensive margins also increase, including the number of convictions and police reports.

These increases in crime are large, but not out of line when compared to previous findings. For instance, Currie and Tekin (2012) find that self-reported childhood maltreatment increases the probability that a young adult reports committing any nondrug crime in the previous 12 months by 8.6 percentage points, relative to 10.9 percent among respondents who report no maltreatment. Our data on the police-reported violence is likely to capture more serious cases, so a large effect is a reasonable finding.

The broader deterioration in outcomes is also visible in other dimensions: Appendix Table 12 shows increases in addiction-related healthcare use, drug convictions, and teenage parenthood among victims.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	GPA (9th grade)	Secondary educ.	HE	Wage earnings	Any MH	Teenage fertility	NEET
Victim	-0.398***	-0.203***	-0.151***	-2739.9***	0.0669***	0.0222***	0.0926***
	(0.0251)	(0.0126)	(0.0201)	(780.6)	(0.00797)	(0.00431)	(0.0251)
Measure age	18	22	25	25	20	19	25
Group	All	All	All	All	All	All	All
N	8760	4472	1058	1058	7176	8118	1058
Victim mean (SD)	6.26 (1.641)	.535 (.499)	.071 (.257)	11995 (13898)	.174 (.379)	.047 (.228)	.44 (.826)
Non-treated mean (SD)	6.638 (1.825)	.731 (.443)	.215 (.411)	14968 (14615)	.106 (.307)	.024 (.168)	.29 (.669)
Controls	Cell FE	Cell FE	Cell FE	Cell FE	Cell FE	Cell FE	Cell FE

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Table 3: Matching estimates for economic, educational, and health outcomes. The sample uses the 1:1 exact-plus-propensity-score match in Table A.3. Outcomes are measured at the ages listed in the table. Regressions include matching-cell and calendar-year fixed effects. Standard errors are clustered by matching cell.

Table 5 shows the estimated effects on the probability of different criminal careers. Victims are much less likely to be classified as having "No recorded offending." This group has no recorded offending throughout the follow-up period. The probability of membership increases in each of the other criminal-career groups, with the largest increase in both absolute and relative terms in the "Persistent offending" category.

	(1) Cum. any conv.	(2) Cum. N conv.	(3) Cum. any rep.	(4) Cum. rep.	(5) Cum. any viol. conv.
Victim	0.144***	1.159***	0.222***	4.663***	0.0740***
	(0.00909)	(0.185)	(0.0118)	(0.552)	(0.00601)
Measure age	21	21	21	21	21
Group	All	All	All	All	All
N	6026	6026	6026	6026	6026
Victim mean (SD)	.232 (.422)	1.737 (9.11)	.591 (.492)	7.373 (26.537)	.1 (.3)
Non-treated mean (SD)	.089 (.284)	.553 (4.996)	.369 (.483)	2.674 (15.619)	.026 (.158)
Controls	Cell FE	Cell FE	Cell FE	Cell FE	Cell FE

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Table 4: Matching estimates for crime outcomes at age 21. The sample uses the 1:1 exact-plus-propensity-score match in Table A.3. Regressions include matching-cell and calendar-year fixed effects. Violent convictions include petty assault, assault, aggravated assault, attempted homicide, and sexual offenses. Standard errors are clustered by matching cell.

	(1) Not offender	(2) Infrequent offender	(3) Adolescent limited	(4) Persistent offender
Effect of of being a victim				
Linear probability model	-0.232***	0.0644***	0.0805***	0.0872***
	(0.0118)	(0.0115)	(0.00947)	(0.00594)
Multinomial logit (AME)	-0.232***	0.0648***	0.0784***	0.0884***
	(0.00228)	(0.00230)	(0.00198)	(0.00150)
Sample mean	0.466	0.300	0.171	0.063
N (Individuals)	1,010	1,010	1,010	1,010

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Matching estimates for criminal careers using a linear probability model and multinomial logit. The LPM estimates use Eq. (2) and match on the exact variables and propensity score listed in Table A.3. The MNL does not use the match-cell fixed effects for computational reasons. All regressions include matching-cell and calendar-year fixed effects. The sample includes victims and controls from the 1:1 match and is restricted to individuals who can be followed through age 25. The table reports linear probability model estimates and average marginal effects from the multinomial logit. Standard errors are clustered at the matching-cell level.

5.1.3 Robustness checks

This section describes some robustness checks we perform to assess the reliability of our matching results.

Results by previous healthcare use

To assess whether the estimates are driven by the detection of previously unrecorded conditions, we repeat the analysis for victims who had a mental-health contact before the first recorded event and their matched controls. Figure A.12 shows similar increases in mental-healthcare use among victims who had already received treatment before the event.

Other simultaneous events

Other family events may occur around the first recorded episode of parental violence. Some may precede or share common causes with the violent event, while others—such as parental conviction or separation—may themselves be consequences of it. Because these events can be mediators, excluding them does not identify the direct effect of violence. We use the restrictions instead as a sensitivity exercise asking whether the main estimates are concentrated among families experiencing particular concurrent changes, as in previous work (e.g. Adams et al., 2026).

We define parental conviction as either parent being convicted of a crime in event years 0 or 1, separation as the child living with both parents at event time -1 but not at event time 1, and job loss as either parent being employed at event time -1 but not at event time 1.

Figure A.16 reports the results. The estimates remain similar across these restrictions, indicating that the main patterns are not concentrated solely among families experiencing these accompanying events.

Different sets of matching variables

We assess whether the long-run estimates change with the variables used in the matching procedure. We compare exact matching on the child's mental-health measures and parental socioeconomic characteristics with exact matching combined with a propensity score based on additional background variables (see Table A.3).

Figure A.14 reports estimates for the main long-run outcomes. This stability shows that the long-run estimates are not sensitive to the particular set of observed characteristics used in the matching procedure.

Additional controls

In a similar fashion to the previous check, we also include additional controls in the post-matching specification. We measure the control variables before the first recorded event. The additional controls include the mother's and father's earnings, employment status, blue-collar employment, healthcare admissions, recorded victimizations, police reports, and crime history; the father's violent victimizations; the child's pre-event specialist-care diagnoses for mental-health conditions, depression, anxiety, intellectual disabilities, developmental disorders, and behavioral or emotional disorders; the mother's and father's specialist-care diagnoses for anxiety, depression, addiction, and behavioral or emotional disorders; their purchases of antidepressants and other nervous-system medication; the mother's primary-care addiction treatment; and the father's benzodiazepine purchases.

The results, reported in Fig. A.14 for secondary education and any criminal convictions, remain quite similar to the ones we obtain from the main specification.

Placebo outcomes

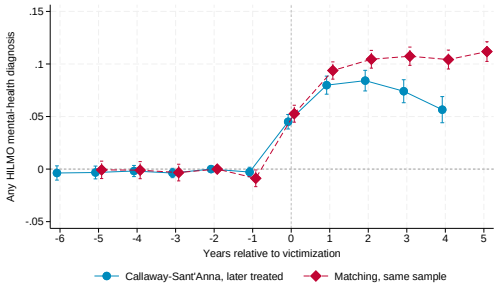
We construct placebo outcomes from the healthcare data: diagnoses of digestive-system disorders, visual or hearing impairments, and metabolic disorders. A discontinuity in these outcomes would raise concern about residual health-related selection or a generalized increase in diagnosis following the event. Figure A.13 reports the results. We find no clear increase around the event, providing no evidence of a broad increase in the detection of unrelated health conditions.

5.2 Comparison to later-treated

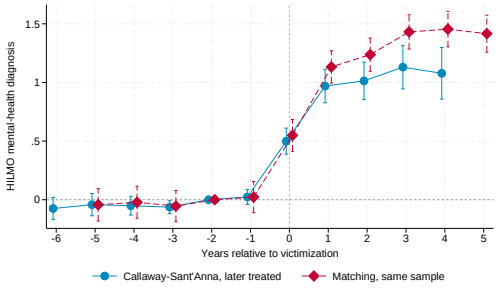
Figure 2 presents estimates using later-treated individuals as controls. The figures show no evidence of differing outcome paths before the first recorded event, followed by a clear increase in recorded mental-health diagnoses and visits at the event. The increase persists throughout the five-year follow-up period.

The estimates are smaller in absolute terms than the matching estimates in Section 5.1. Our main explanation is the difference in samples. Because the later-treated comparison requires eligible later-treated individuals to act as controls, the sample is skewed toward younger children. Among younger children, both baseline mental-healthcare use and the post-event increases are also smaller in the matching design.

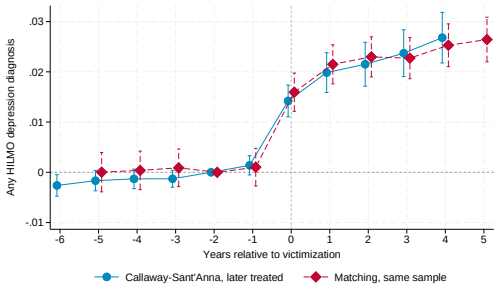
To show this, Fig. 2 also presents matching estimates for the same group. The estimates differ in some later periods, but the differences have no clear direction and are small relative to the initial increases. The exercise does not make the estimates identical by construction. The similarity is reassuring because the two designs rely on different identifying assumptions and different comparison groups.



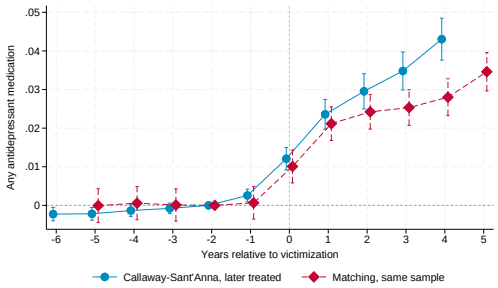
(a) Any mental health diagnosis



(b) N of specialist visits with MH diagnosis.



(c) Estimates: Any mood disorder diagnosis



(d) Any antidepressant prescription

Figure 2: Estimates using later-treated individuals as the comparison group and the method of Callaway and Sant'Anna (2021). The figure also includes matching estimates for the same group.

5.3 Sibling results

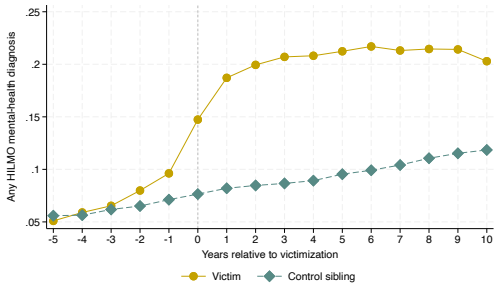
5.3.1 Effects in childhood

Recorded victims have somewhat higher rates of mental-health treatment than comparison siblings before the event, but the pre-event changes are similar. The difference increases sharply in the event year and remains elevated, although it declines later in the follow-up period. From two years before the event to the following year, the gap in any specialized-care mental-health diagnosis widens by 8.8 percentage points in the sibling comparison and by 9.5 percentage points in the matching design.

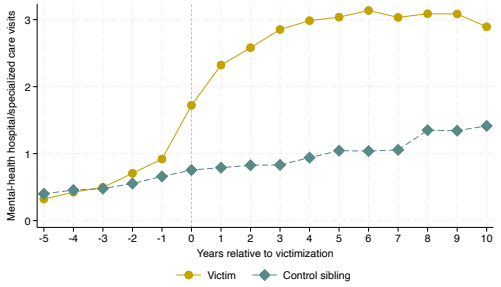
The appendix reports sibling estimates for antidepressant use and primary-care mental-health contacts in Fig. A.4. These outcomes also increase after the recorded event.

Comparison siblings have higher pre-event rates of several adverse outcomes than matched controls, so the 8.8 percentage-point increase is also smaller relative to their baseline. This difference is consistent with siblings sharing family circumstances that are not fully captured by the matching variables.

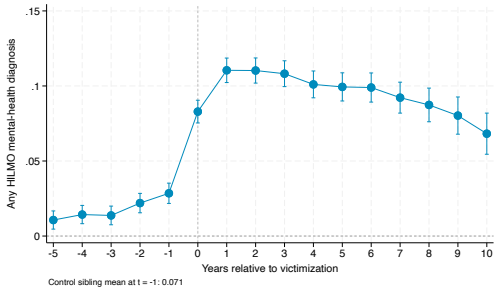
As discussed in the methods section, several factors could explain the smaller sibling estimates. Mother fixed effects remove persistent family characteristics that remain between matched families, while unreported violence against a sibling who is classified as a comparison sibling would tend to reduce the within-family estimate. The two designs also estimate different effects. The sibling comparison captures components of the event that affect the recorded victim and sibling differently. Family separation, child-protection involvement, or a parent's conviction is differenced out only when its effect is the same for both siblings; differences in the effects of these changes remain in the estimate.



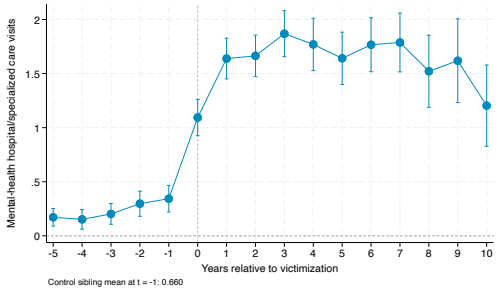
(a) Raw means: Any mental-health diagnosis



(b) Raw means: Number of specialized-care visits with a mental-health diagnosis



(c) Estimates: Any mental-health diagnosis



(d) Estimates: Number of specialized-care visits with a mental-health diagnosis

Figure 3: Sibling raw means and event-study estimates using Eq. (3). Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event mean for comparison siblings.

	(1) GPA (9th grade)	(2) Secondary educ.	(3) HE	(4) NEET	(5) Wage earnings	(6) Any MH
Victim	-0.132***	-0.143***	-0.0372***	0.0655***	-3234.9***	0.0665***
	(0.0271)	(0.0126)	(0.00909)	(0.0150)	(470.1)	(0.00780)
Measure age	18	22	25	25	25	20
Group	All	All	All	All	All	All
N	11047	9893	7675	7675	7675	14902
Victim mean (SD)	6.065 (1.696)	.586 (.493)	.064 (.245)	.257 (.437)	11370.487 (14350.713)	.169 (.375)
Non-treated mean (SD)	6.493 (1.657)	.675 (.469)	.093 (.29)	.244 (.429)	13820.887 (14772.358)	.091 (.288)
Controls	Mother FE	Mother FE	Mother FE	Mother FE	Mother FE	Mother FE

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Table 6: Sibling estimates for economic, educational, and health outcomes using Eq. (4). Outcomes are measured at the ages listed in the table. Standard errors are clustered by mother.

5.3.2 Effects in adulthood

The matching and sibling estimates are similar for secondary-education completion, earnings, and criminal convictions. This agreement is difficult to reconcile with an explanation based solely on persistent family-level disadvantage, because the sibling specification compares children within the same family. While it is consistent with a large role for the recorded event itself, it does not establish that the estimates reflect the violent act alone. Differential changes within the family, including differences in child-protection involvement, living arrangements, or responses to a parent's conviction, can affect the recorded victim and sibling differently and remain in the sibling estimate. The adult results therefore rest on stronger assumptions than the short-run results, for which the matching, later-treated, and sibling comparisons also show the timing of the outcome change.

	(1) Cum. any conv.	(2) Cum. N conv.	(3) Cum. any rep.	(4) Cum. rep.	(5) Cum. any viol. conv.
Victim	0.0980***	0.938***	0.123***	3.588***	0.0507***
	(0.00969)	(0.238)	(0.0113)	(0.653)	(0.00685)
Measure age	21	21	21	21	21
Group	All	All	All	All	All
N	13236	13236	13236	13236	13236
Victim mean (SD)	.228 (.419)	1.774 (10.098)	.556 (.496)	7.461 (26.789)	.093 (.291)
Non-treated mean (SD)	.158 (.365)	1.098 (7.198)	.451 (.498)	4.413 (19.586)	.062 (.24)
Controls	Mother FE	Mother FE	Mother FE	Mother FE	Mother FE

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Table 7: Sibling estimates for convictions and police reports at age 21. "Cum. any conv." is the cumulative probability of any conviction, and "Cum. any rep." is the cumulative probability of any police report. Violent convictions include petty assault, assault, aggravated assault, attempted homicide, and sexual offenses. Standard errors are clustered by mother.

5.3.3 Robustness checks

The sibling checks address the possibility that recorded victims would have had worse adult outcomes than their siblings even without the recorded event.

Full siblings

In our main specification, mother fixed effects compare siblings who share a mother. Full siblings may share more of their childhood environment. Table A.6 adds mother-by-father fixed effects and yields a similar estimate.

Limiting the age difference between siblings

Siblings born many years apart may face different family circumstances. We therefore restrict comparison siblings to those born within four years of the recorded victim. Requiring the same birth year leaves too few observations for precise estimates.

Tables A.6 and A.7 report the adult estimates when comparison siblings must be within four years of the victim's age. The magnitudes remain similar to those in the main sibling specification.

Additional covariates

The pre-event outcome means show that recorded victims have worse health than their comparison siblings on some measures. The event studies display these level differences directly, but pre-event health could also predict worse adult outcomes in the absence of recorded victimization. This would bias the long-run sibling estimates away from zero.

We observe the siblings before the event and adjust for pre-event health in Tables A.6 and A.7. The estimates change little. We also restrict the sample to victims with no mental-health contact before the event. The estimates remain similar (Tables A.6 and A.7). Further robustness checks on different outcomes can be found in Fig. A.15.

6 Heterogeneity and effects on parents

6.1 Heterogeneity

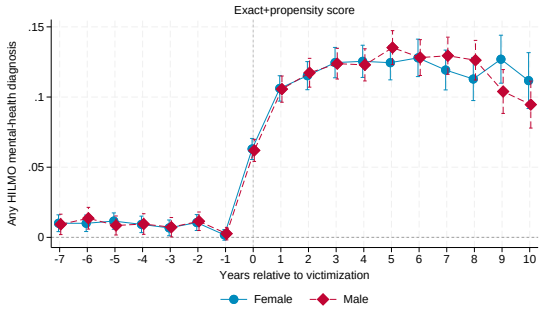
We explore heterogeneity along two main dimensions. First, because boys and girls often have different recorded mental-health diagnoses and propensities to offend, we estimate the effects separately by gender. Theories in economics, psychology, and other disciplines

predict different responses depending on the age at which a shock occurs. We therefore also estimate the effects separately by age at first recorded victimization.

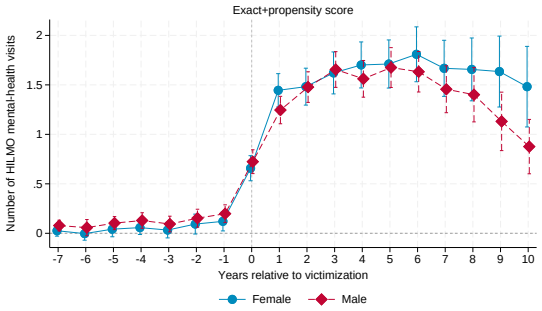
We find similar absolute increases in the overall mental-health outcome for boys and girls. The composition of recorded diagnoses differs substantially: Fig. A.11 shows that the largest increases for girls are in mood and anxiety diagnoses, while boys have larger increases in recorded ADHD and conduct-disorder diagnoses (ICD-10 F9). Thus, the overall increase in mental-health treatment is not concentrated in one gender, but the recorded diagnoses differ.

The dynamics also vary with age at first recorded victimization. Mental-health treatment increases following the recorded event in all age groups. For children first recorded as victims at younger ages, the initial increase is smaller in absolute terms but remains elevated for longer. Among adolescents, the increase around the event is considerably sharper, followed by a gradual decline. Relative to the much lower baseline prevalence of mental-health treatment at younger ages, the increase is particularly large for younger children. The dynamics differ substantially by age at first recorded victimization, both in the magnitude of the immediate response and in its persistence.

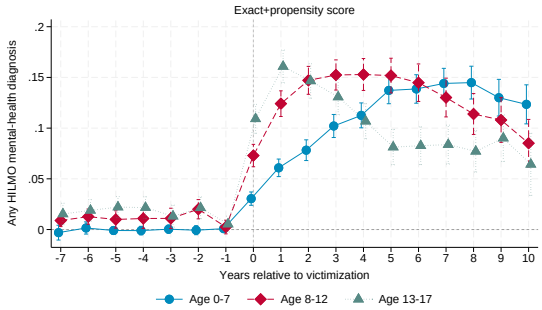
We report heterogeneity in the adult outcomes in the Appendix. The decline in secondary-education completion is similar for boys and girls, while the increase in criminal convictions is substantially larger for boys. The point estimates also suggest larger increases in adult crime among children first recorded as victims at older ages. These subgroup estimates should be interpreted as descriptive evidence of heterogeneity; in particular, differences in point estimates do not by themselves establish statistically significant differences across groups. Figures A.7 and A.8 report a set of estimates by gender, age at victimization, maternal education, identity of the suspected perpetrator, and whether a physical injury is recorded in the event year.



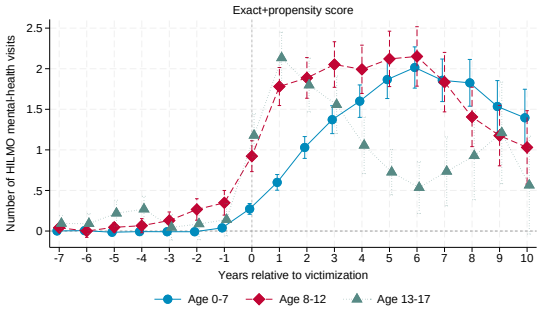
(a) Any mental-health visit



(b) Number of mental-health visits



(c) Any mental-health visit



(d) Number of mental-health visits

Figure 4: Effects on mental-health visits by gender and age group

6.2 Effects on parents

We also use the matching approach to examine changes in parents' outcomes around the first recorded event. These outcomes are of independent interest, provide information about the events, and help assess possible mechanisms.

Figure A.10 shows changes in parents' outcomes around the recorded event. Parental labor-market outcomes deteriorate around and after the recorded event, particularly for suspected perpetrators. These changes may reflect criminal-justice consequences, family disruption, or other changes associated with the event. One plausible interpretation is that the larger deterioration among suspected perpetrators reflects criminal-justice consequences and other changes that follow the report.

Parents in families with recorded violence already have lower employment and earnings and greater mental-healthcare use before the event. These pre-existing differences underscore the selection problem motivating the matching design. Figure A.14 shows that adding these pre-event parental characteristics as controls leaves the main long-run estimates largely unchanged.

Second, a negative parental labor-market shock could both trigger the first recorded event and cause subsequent changes in mental-healthcare use. To examine this possibility, we restrict the sample to families without sudden labor-market changes around the event. The results remain similar (Fig. A.16).

7 Conclusions

We examine the effects of violence by parents on children's outcomes in both the short and long run. Although parental violence is widely recognized as a severe adverse childhood experience, little previous evidence uses large-scale administrative data and accounts for differences that precede exposure. We find evidence that exposure to parental violence has lasting consequences for victims. Beyond immediate and persistent increases in mental-healthcare use, the estimates indicate lower educational attainment and increased adult criminal offending. We find increases in both adolescent-limited and life-course-persistent offending.

The short-run increase appears in the matching, later-treated, and sibling comparisons. The adult estimates rely on matching and sibling comparisons and therefore require stronger assumptions. Their similarity is consistent with substantial negative effects of violence itself, but it does not isolate the violent act from all other changes that follow the report. The sibling estimate captures the components of the event that affect the recorded victim and sibling differently: common family consequences are differenced out only when they affect both children equally, while differences in their effects remain. The main story in our results —that there are sizable negative consequences for the victimized children — is consistent throughout different designs with differing assumptions.

Previous results suggest that the children who are victimized fare poorly later in life. Our results suggest that these findings are unlikely to solely reflect persistent family-level disadvantage. These results motivate further work on which responses after a report—including healthcare, child protection, and changes in living arrangements—can alleviate the lasting consequences for children.

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Appendix to

The Short- and Long-Run Consequences of Parental Violence Against Children

Pedro Carneiro, Ossi Tahvonen

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A.1 Data and methods

Table A.1: Healthcare categories and corresponding ICD-10 codes

Subcategory	**ICD-10 codes**
Mood and depressive disorders	F3
Anxiety disorders	F4
Behavioural and emotional disorders with onset usually occurring in childhood and adolescence ("Conduct/ADHD")	F9
Intellectual disability	F7
Developmental disorders	F8
Any mental health disorder	Any F chapter code
Physical injury	Any S chapter code

Notes: ICD-10 codes are reported at the chapter/block-prefix level. The category "Any mental health disorder" includes any ICD-10 diagnosis code beginning with F.

Table A.2: Violent crime categories and corresponding offence codes

Subcategory	**Crime codes**
Sexual offences	20010, 20020, 20030, 20040, 20050, 20060, 20070, 20080, 20090, 20120, 20130, 20140, 20150, 20160, 20180
Assault and attempted homicide	21010, 21020, 21030, 21040, 21050, 21060, 21070, 21120, 21130

Notes: Crime codes indicating violent victimization, using Statistic Finland crime codes 2023. Codes are reported as five-character prefixes. Excluded from our data are completed homicide crimes (since the victim cannot be followed), terrorism offences, and other rare and exceptional crime categories (for privacy reasons). Completed homicide offences are excluded from the sample.

Table A.3: Matching specifications, variables, and matching rates

Specification	Matching variables	Matching rate	Use
Age and gender	Event year; child age (two-year categories) and gender; mother's and father's age (five-year categories)	100% (1:3)	Descriptive benchmark
Exact	Age-and-gender variables; mother's and father's education; father born in Finland; father in a blue-collar occupation; child mental-health contact; mother's mental-health medication; father's conviction; child living with mother; mother's employment history	98.3% (1:3)	Childhood event studies
Exact + propensity score	Exact variables; within-cell propensity-score match	92.2% (1:1)	Main adult estimates; childhood comparison

Notes: The matching rate is the percentage of otherwise eligible victims with the required number of controls. The exact-plus-propensity-score specification used for the estimates in Tables A.5, 3 and 4. All variables are measured before the first police-recorded event. Unmatched victims are excluded.

The propensity score is estimated with linear regression using the following variables:

- Mother's and father's mental-health medication purchases during the three years before the event
- Mother's and father's criminal convictions during the three years before the event
- Mother's criminal conviction in the year before the event
- Whether the child lived with the mother and father in the year before the event
- Mother's earnings decile
- Calendar-year indicator
- Mother's receipt of social assistance
- Child diagnoses of developmental disorders during the two years before the event
- Child diagnoses of intellectual disability during the two years before the event
- Child diagnoses of behavioral or emotional disorders with onset in childhood or adolescence during the two years before the event
- Number of the child's specialist-care mental-health visits in the year before the event
- Mother's and father's addiction-related healthcare during the three years before the event
- Mother's recorded victimization during the three years before the event
- Mother's and father's employment histories during the five years before the event
- Father's social-assistance history during the five years before the event
- Whether either parent had a violent-crime conviction during the two years before the event
- Mother's and father's marital status
- Mother's and father's employment status
- Mother's and father's earnings

A.1.1 Sequence analysis

This appendix describes the sequence-analysis procedure used to construct the criminal-career groups discussed in Section 3.8.

Sequences and states. We construct an annual criminal-history sequence for each individual covering ages 15 through 25. The sequence therefore contains eleven yearly observations. In each year, an individual is assigned to one of two states:

$$s_{ia} = \begin{cases} 0 & \text{if individual } i \text{ has no recorded crime at age } a, \\ 1 & \text{if individual } i \text{ is suspected or convicted of at least one crime at age } a. \end{cases}$$

We include only individuals who can be observed from ages 15 through 25 and whose victimization event, or corresponding event date assigned to a control, occurs before age 16.

Sequence distance. We measure the dissimilarity between the criminal histories of two individuals using optimal matching. The procedure assigns a cost to transforming one individual's sequence into another through substitutions and insertions or deletions. We use an insertion/deletion cost of 5 and substitution costs of 1. Thus, two individuals have a small distance when criminal involvement occurs at similar ages and a larger distance when their trajectories differ substantially in either the timing or persistence of offending.

Clustering. We use the resulting pairwise distance matrix to group individuals with similar criminal trajectories. We apply Ward's hierarchical clustering to the distance matrix. We select a four-cluster solution based on interpretability.

The four-cluster solution produces groups with clearly different age profiles of recorded offending. Cluster 1 contains individuals with no recorded offending during ages 15–25 and is labelled "No recorded offending." Cluster 2 contains individuals with relatively infrequent offending spread across the observation period and is labelled "Infrequent offending." Cluster 3 has offending concentrated at younger ages followed by a substantial decline and is labelled "Adolescent-limited offending." Cluster 4 has persistently high rates of recorded offending and is labelled "Persistent offending." Figure A.3 plots, for each cluster, the share of individuals recorded as offending at each age.

The labels are intended to summarize the empirical trajectories rather than impose ex ante criminological categories on the data. In particular, the administrative measure combines police suspicion and convictions and covers only ages 15–25, with no information on earlier age behaviour. It also captures only reported crime. The resulting clusters should therefore not be interpreted as direct empirical counterparts to the life-course categories defined in Moffitt (1993). They instead provide a data-driven summary of heterogeneity in the timing and persistence of recorded criminal behaviour.

Cluster membership as an outcome. In the main analysis, we treat membership in each of the four mutually exclusive clusters as an outcome. We estimate differences between victims and matched controls in the probability of belonging to each cluster using both linear probability models and a multinomial logit model. For the multinomial logit specification, we report average marginal effects so that the estimates can be interpreted as changes in the probability of membership in each criminal-career group.

A.2 Descriptive statistics and results

	(1) Gen. pop. mean	(2) Matched control mean	(3) Victims mean
Age at measurement	7.86	7.90	7.89
Child male	0.53	0.53	0.53
Mother Age	36.69	36.51	36.26
Father Age	39.58	39.63	39.66
Father secondary	0.56	0.55	0.55
Father HE	0.25	0.14	0.14
Mother urban area	0.84	0.88	0.88
Mother married	0.23	0.24	0.24
Mother homeown	0.72	0.50	0.50
Mother born in Fin	0.90	0.80	0.80
Father born in Fin	0.90	0.79	0.79
Father blue collar	0.38	0.21	0.21
Mother any MH drug 3y	0.15	0.24	0.24
Any MH contact (drug/healthcare vis.)	0.05	0.12	0.12
Any MH diagnosis	0.04	0.08	0.09
Father any MH drug 3y	0.10	0.12	0.17
Father conv. 3y	0.04	0.06	0.10
Lives w Father	0.79	0.70	0.62
Mother wage earnings	26058	23475	21480
Father wage earnings	40819	34906	32953
Mother employed	0.74	0.64	0.56
Father empl.	0.84	0.74	0.69
Mother Unemployed	0.08	0.12	0.16
Father Unemployed	0.09	0.14	0.18
Mother convicted in the last 3y	0.01	0.02	0.05
Any MH diagnosis	0.04	0.08	0.09
HC Any develop. dis.	0.02	0.04	0.04
HC Any beh/att. dis.	0.02	0.04	0.05
Prim. care MH visits	0.06	0.12	0.11
BW 1.5-2 kg	0.00	0.00	0.01
BW 2-2.5 kg	0.01	0.01	0.01
Preterm	0.03	0.03	0.03
Sepsis	0.02	0.02	0.02
Other defect	0.01	0.01	0.01
Asyxia	0.01	0.01	0.01
Breathing	0.04	0.04	0.05
Circulation	0.04	0.04	0.04
Hosp. visits	1.29	1.80	2.29
Any MH diagnosis	0.04	0.08	0.09
Any anx. diag.	0.01	0.01	0.01
Any Depr. diag.	0.00	0.01	0.01
HC Endocr.	0.05	0.08	0.09
HC Nervous.	0.04	0.07	0.05
HC Musculosc.	0.04	0.05	0.06
Antidepr. presc.	0.02	0.04	0.04
N diff prescr.	1.21	1.27	1.30
Prim. care visits	8.21	8.95	10.38
Single adult HH	0.17	0.25	0.36
Lives w Mother	0.97	0.96	0.92
Mother blue collar	0.58	0.46	0.39
Mother DI	0.01	0.02	0.03
Father DI	0.02	0.03	0.04
Mother secondary	0.53	0.54	0.54
Mother HE	0.33	0.24	0.19
Father HE	0.25	0.14	0.14
M youngest child age	5.58	5.65	5.21
Victim of violence	0.00	0.01	0.02
Observations	58826	56517	18839

Table A.4: Summary statistics of the victim children, matched control group (using the Exact matching) and a similarly aged sample of general population.

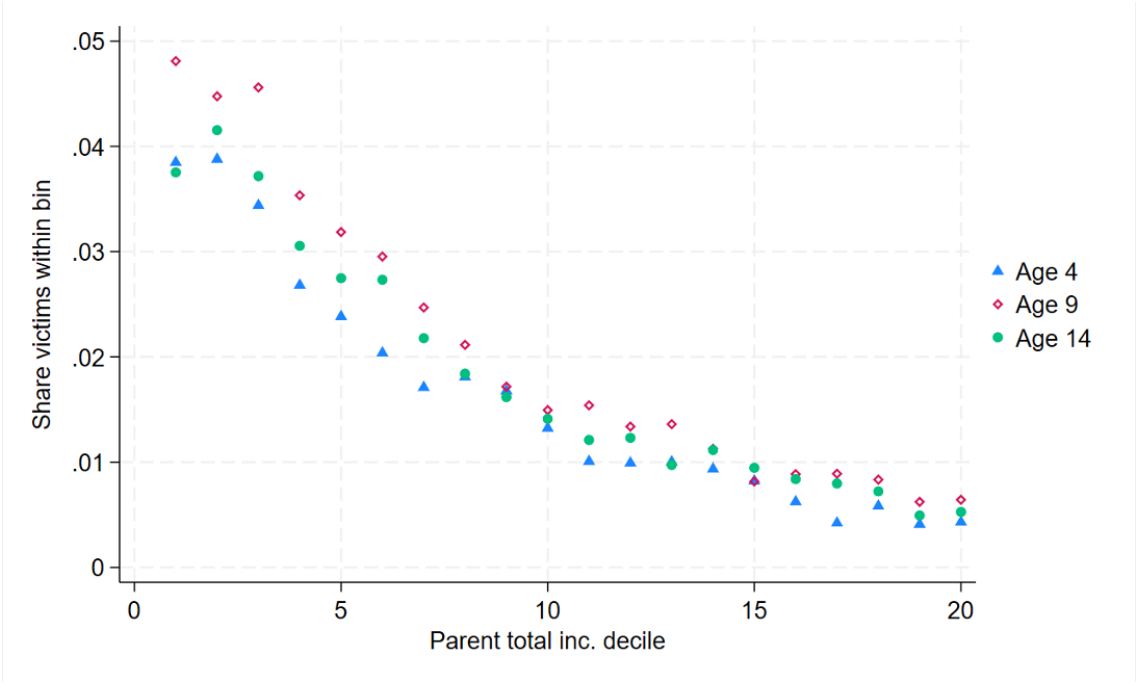


Figure A.1: Victimization by family income. In the x-axis, parental taxable income grouped into 20 quintiles. Y-axis value is the share of children reported as victims in each of the 3 ages. Data from years 2015-2017 only.

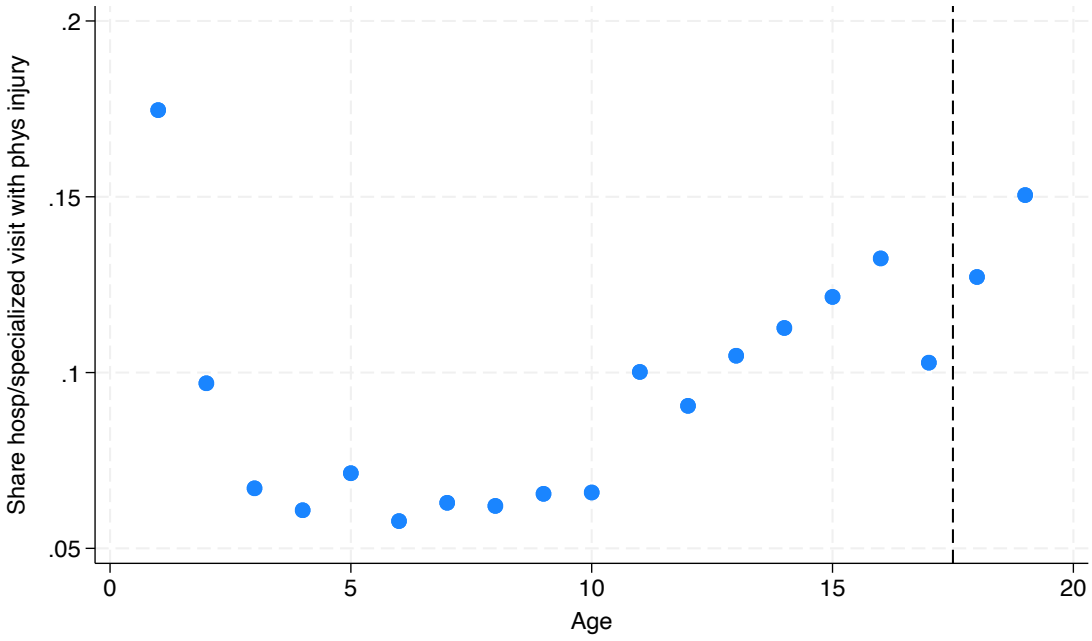


Figure A.2: Share of victims in our sample that visit the hospital/specialized care with physical injuries during the same year as the first victimization event. Physical injury is defined as a visit with a diagnosis from the ICD10 chapter S.

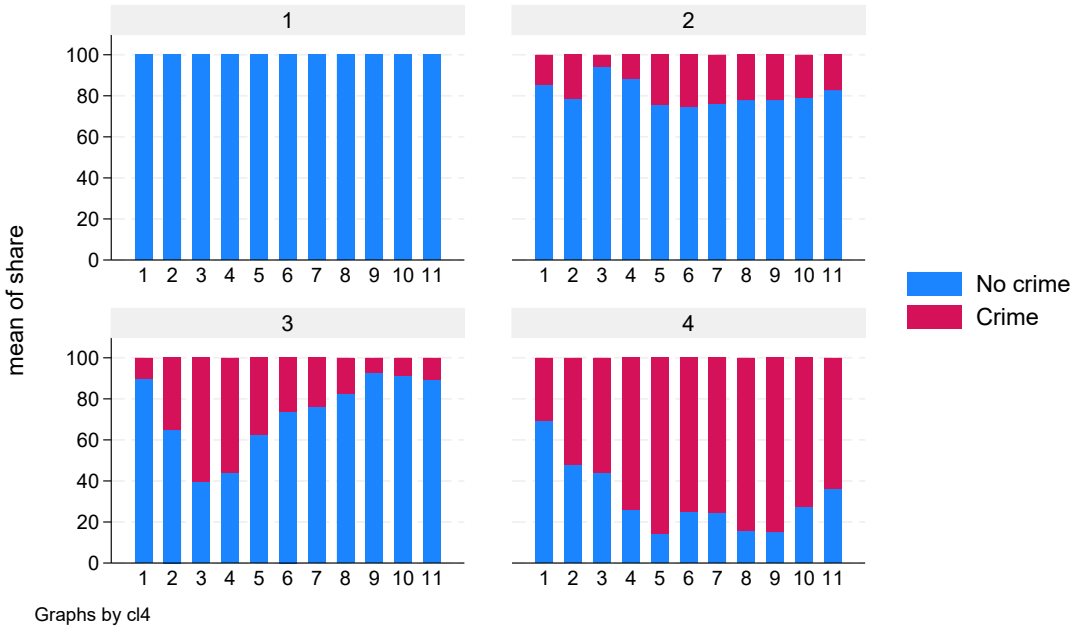
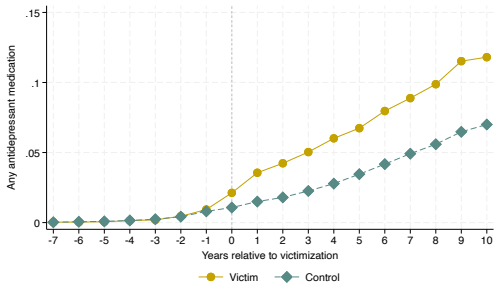
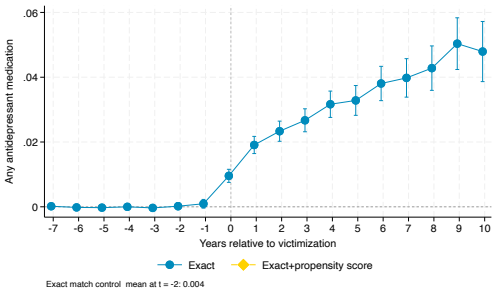


Figure A.3: Percentage of individuals committing crime in the different clusters we identify based on the sequence analysis. The x-axis is years from age 15. Crime=1 if the individual was suspected or convicted of crime during that year For more details, see Section 3.8.

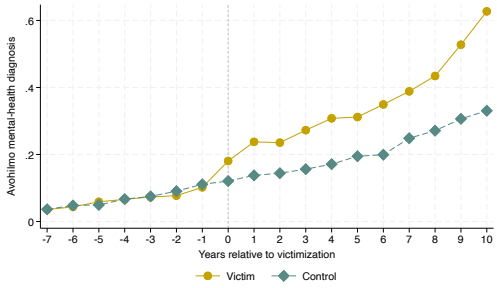
A.3 Additional results



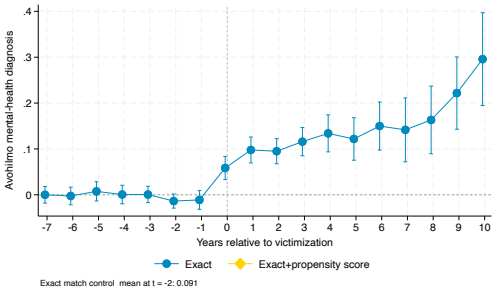
(a) Raw means: any antidepressant



(b) Event study: any antidepressant

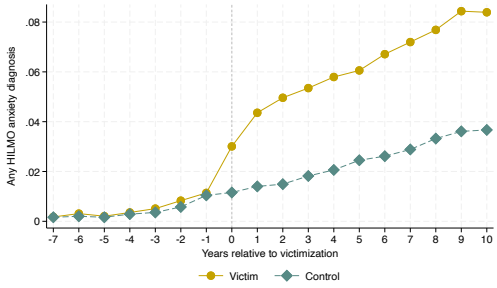


(c) Raw means: Primary care visits for mental health reasons

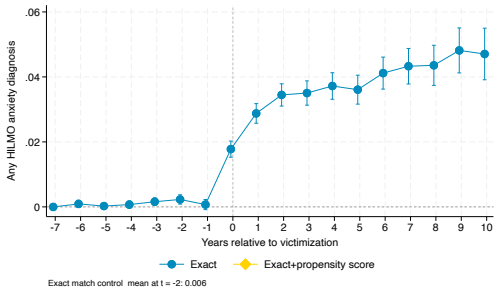


(d) Event study: Primary care visits for mental health reasons

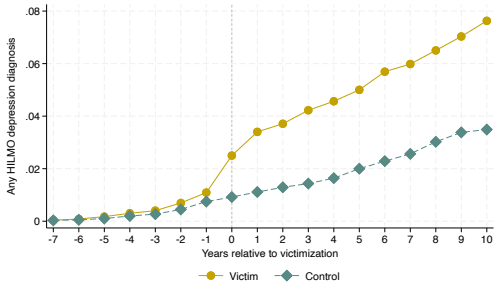
Figure A.4: Matching event-study estimates using Eq. (1) for antidepressant purchases and primary-care mental-health contacts. Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event control mean.



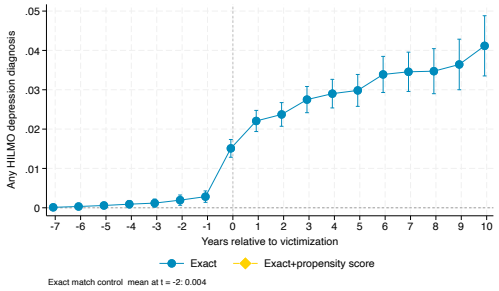
(a) Raw means: Any anxiety disorder (F4) visit



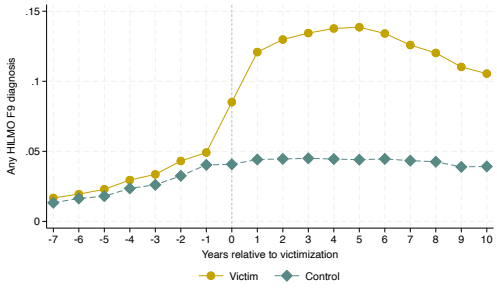
(b) Event study: Any anxiety-disorder diagnosis (F4)



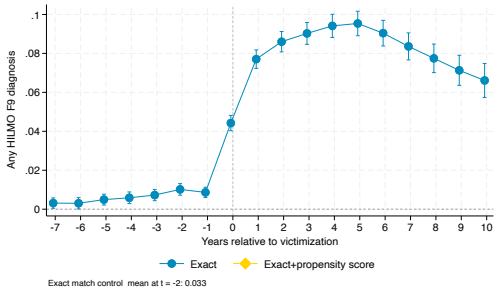
(c) Raw means: Any mood-disorder diagnosis (F3)



(d) Event study: Any mood-disorder diagnosis (F3)

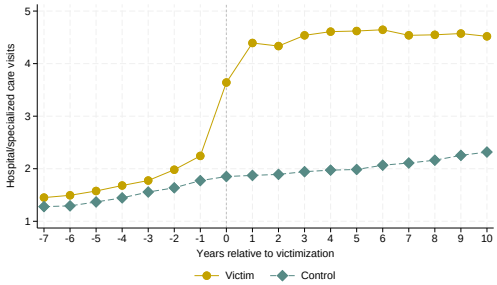


(e) Raw means: Any behavioral or emotional disorder diagnosis (F9)

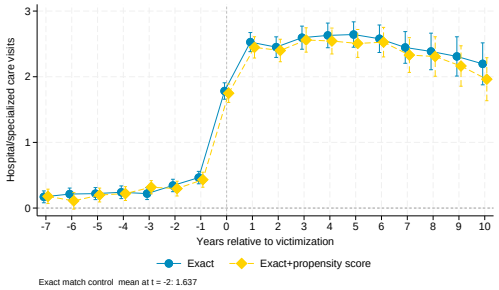


(f) Event study: Any behavioral or emotional disorder diagnosis (F9)

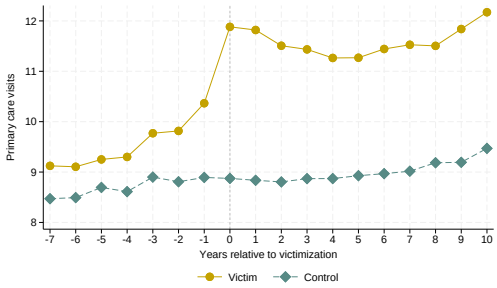
Figure A.5: Matching event-study estimates using Eq. (1) for anxiety, mood, and behavioral or emotional disorder diagnoses. Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event control mean.



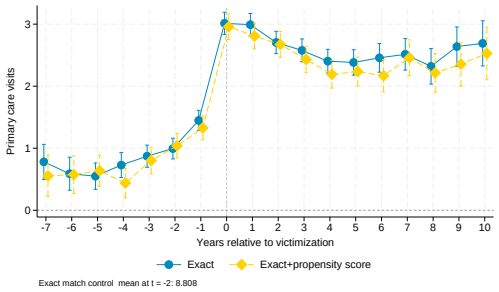
(a) Raw means: Number of hospital/specialized care visits



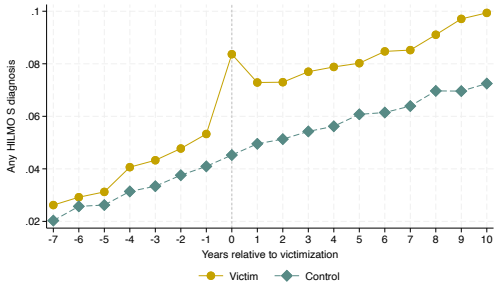
(b) Event study: Number of hospital/specialized care visits



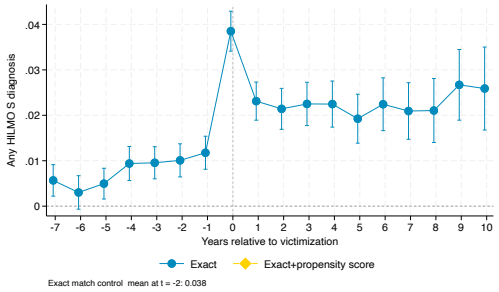
(c) Raw means: Number of primary care visits



(d) Event study: Number of primary care visits



(e) Raw means: Any Physical Injury (S*) visit



(f) Event study: Any Physical Injury (S*) visit

Figure A.6: Matching event-study estimates using Eq. (1) for overall healthcare use and physical injuries. Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event control mean.

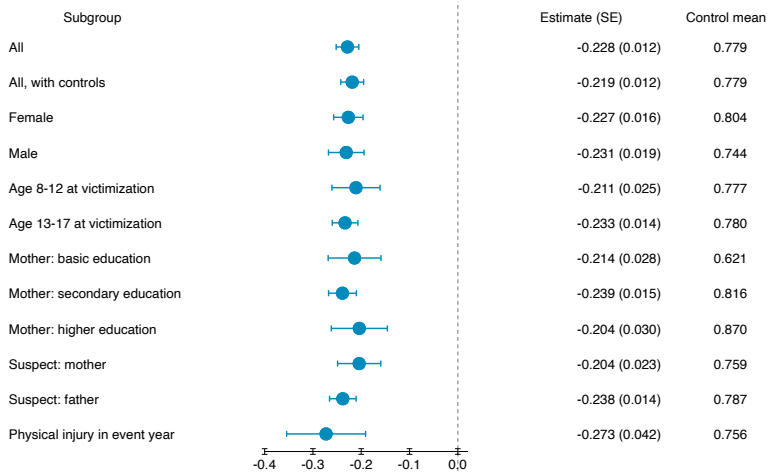


Figure A.7: Heterogeneity in effects on secondary education graduation using the matching design.

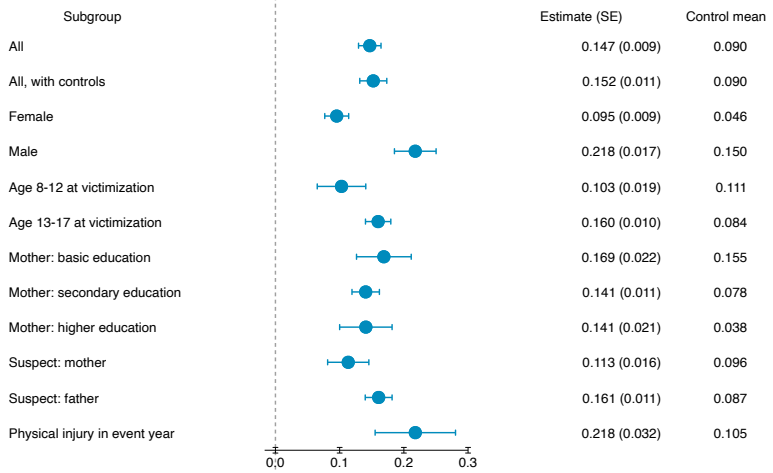
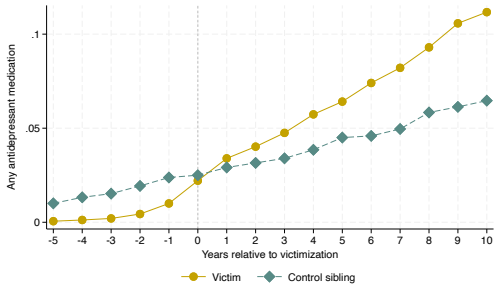
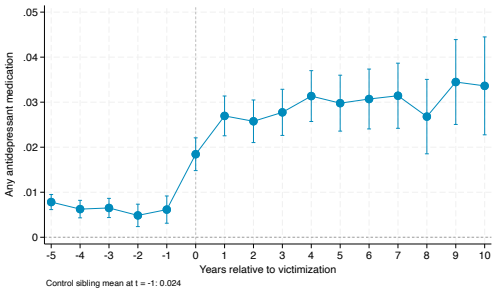


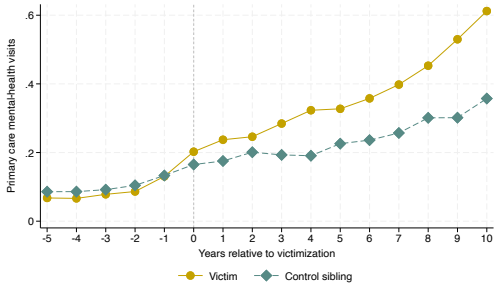
Figure A.8: Heterogeneity in effects on cumulative probability of any conviction using the matching design.



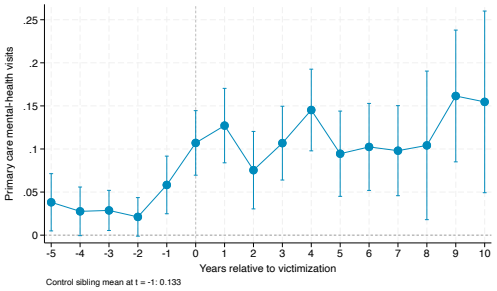
(a) Raw means: any antidepressant



(b) Event study: any antidepressant



(c) Raw means: Primary care visits for mental health reasons



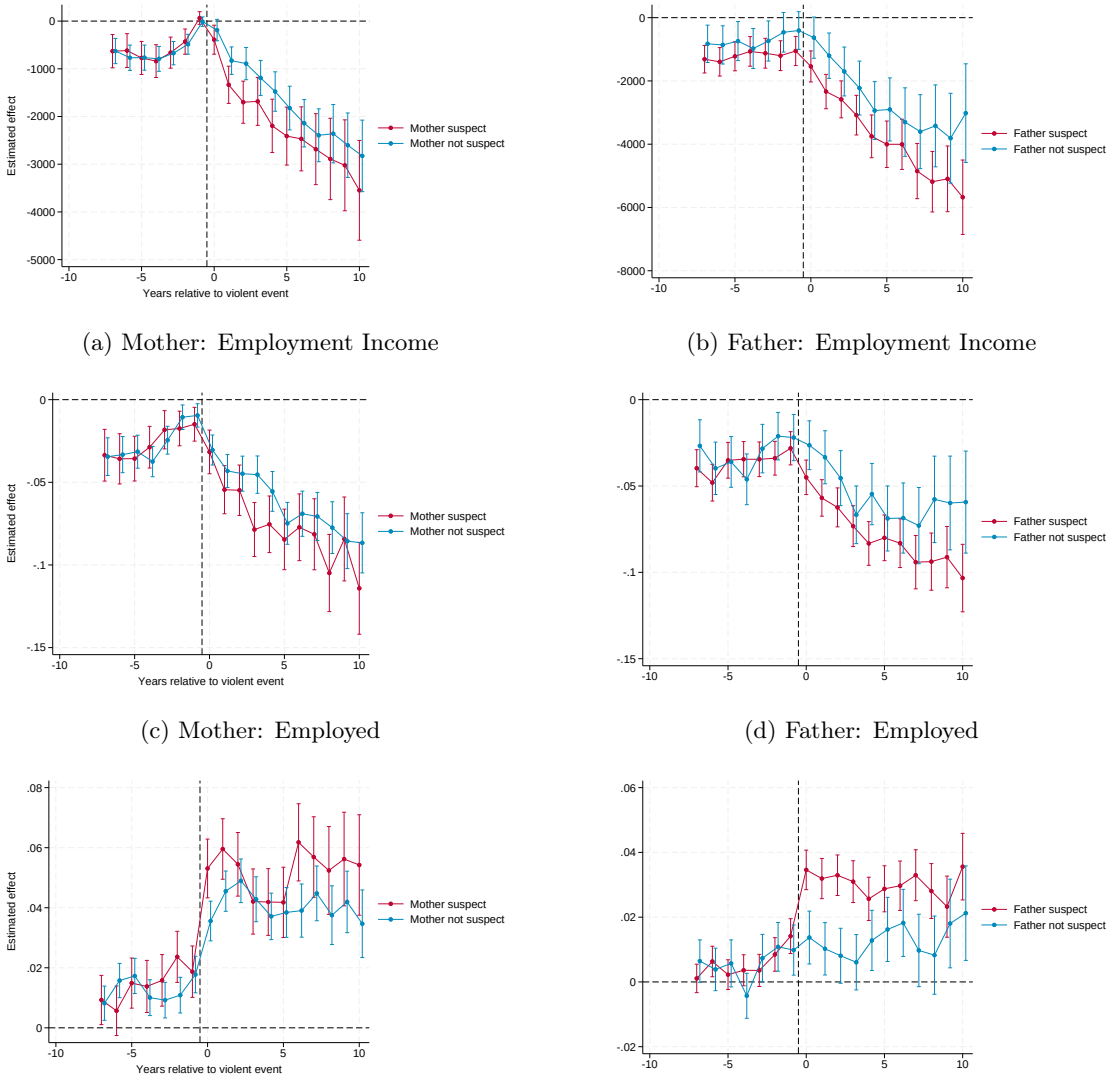
(d) Event study: Primary care visits for mental health reasons

Figure A.9: Sibling event-study estimates using Eq. (3) for antidepressant purchases and primary-care mental-health contacts. Event time is measured in years, and the estimates are in outcome units. Each panel reports the pre-event comparison-sibling mean.

	(1) Any addiction visit	(2) Cum. any drug conv.	(3) Cum. N. drug conv.	(4) Teenage fertility
Victim	0.0275***	0.0461***	0.136***	0.0222***
	(0.00301)	(0.00520)	(0.0197)	(0.00431)
Measure age	18	21	21	19
Group	All	All	All	All
N	10822	6026	6026	8118
Victim mean	.038 (.192)	.067 (.25)	.187 (1.009)	.047 (.228)
Non-treated mean	.011 (.106)	.022 (.146)	.054 (.507)	.024 (.168)
Controls	Cell FE	Cell FE	Cell FE	Cell FE

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Table A.5: Matching estimates for addiction-related healthcare, drug convictions, and teenage parenthood. The sample uses the 1:1 exact-plus-propensity-score match in Table A.3. Regressions include matching-cell and calendar-year fixed effects. Standard errors are clustered by matching cell.



(a) Mother: Employment Income

(b) Father: Employment Income

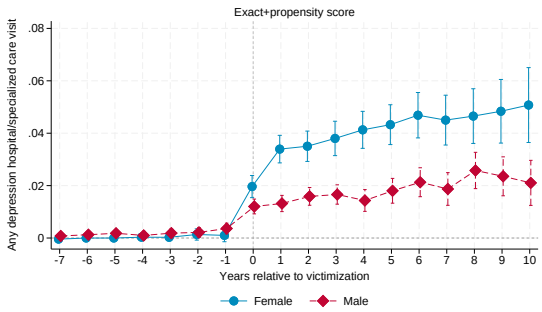
(c) Mother: Employed

(d) Father: Employed

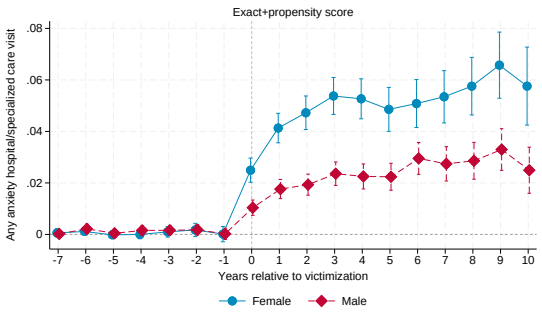
(e) Mother: Any Mental Health diagnosis

(f) Father: Any Mental Health diagnosis

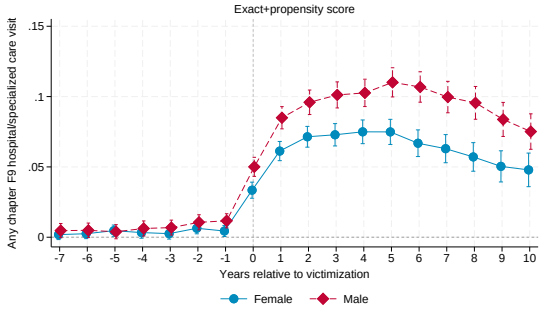
Figure A.10: Parental outcomes around the child's first event. X-axis shows the time, in years, while y-axis shows the estimates using Eq. (1) and Exact+Propensity score matching.



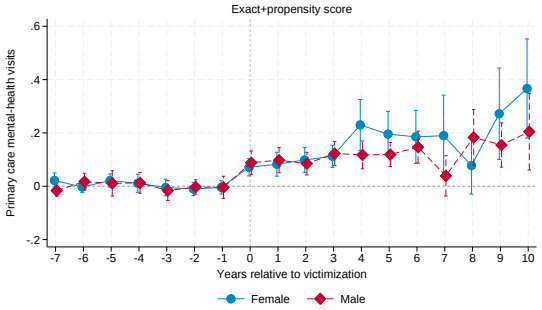
(a) Any HILMO depression diagnosis



(b) Any HILMO anxiety diagnosis



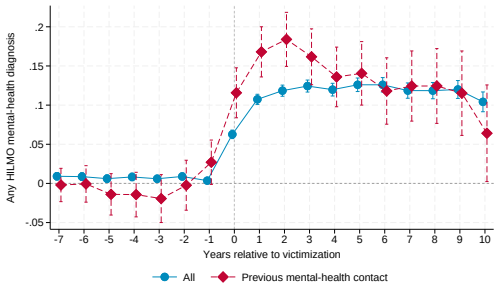
(c) Any HILMO F9 diagnosis



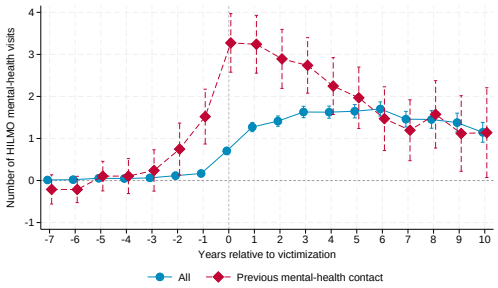
(d) Primary-care mental-health visits

Figure A.11: Effects on mental-health diagnoses and primary-care mental-health visits by gender

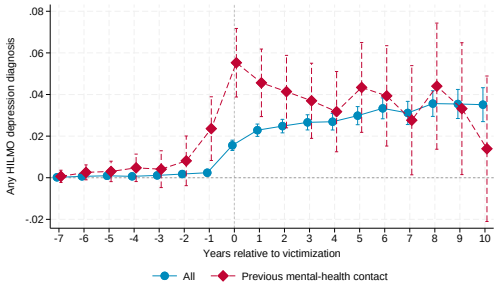
A.4 Robustness



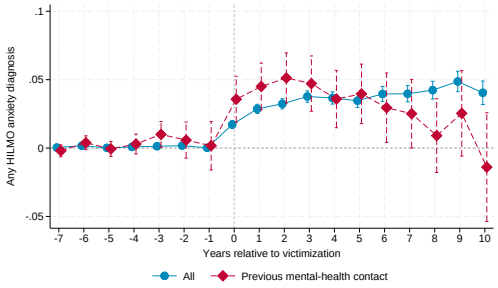
(a) Any mental health diagnosis



(b) N of specialist visits with MH diagnosis.



(c) Estimates: Any mood disorder diagnosis



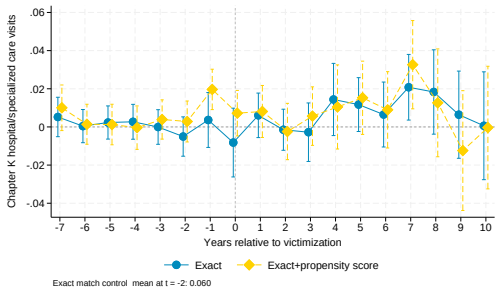
(d) Any anxiety diagnosis

Figure A.12: Event study figures with sample limited to those with previous mental health services contact.

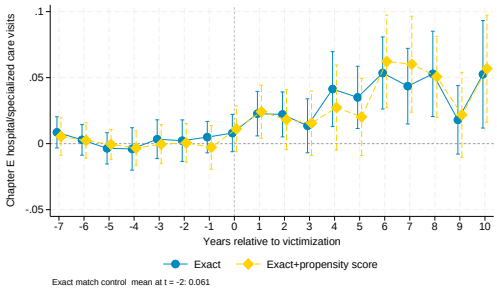
	(1) Secondary educ.	(2) Secondary educ.	(3) Secondary educ.	(4) Secondary educ.	(5) Secondary educ.
Victim	-0.143***	-0.140***	-0.144***	-0.143***	-0.141***
	(0.0126)	(0.0126)	(0.0137)	(0.0146)	(0.0136)
Measure age	22	22	22	22	22
Group	All	All	All	All	All
N	9893	9893	8354	6934	7562
Victim mean (SD)	.586 (.493)	.586 (.493)	.586 (.493)	.588 (.492)	.594 (.491)
Non-treated mean (SD)	.675 (.469)	.675 (.469)	.675 (.469)	.676 (.468)	.690 (.463)
Controls	Mother FE	+health controls	Fam FE	Age diff < 5	No MH pre

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

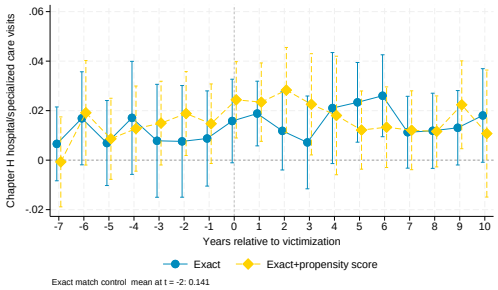
Table A.6: Sibling estimates for secondary-education completion under alternative specifications. Column 2 adds pre-event healthcare controls, column 3 includes mother-by-father fixed effects, column 4 restricts the sibling age difference to four years or less, and column 5 restricts the sample to children with no pre-event mental-health diagnosis. Reported means and standard deviations refer to the victim sample.



(a) Specialist-care visits for diseases of the digestive system



(b) Specialist-care visits for endocrine, nutritional, and metabolic diseases



(c) Specialist-care visits for diseases of the eye, ear, and related structures

Figure A.13: Event-study estimates for placebo health outcomes. Panel (a) reports whether the individual received a diabetes diagnosis. Panels (b)–(d) report the annual number of specialist-care visits with diagnoses in ICD-10 chapters K, E, and H, respectively. The figures compare exact matching with exact matching combined with propensity-score matching.

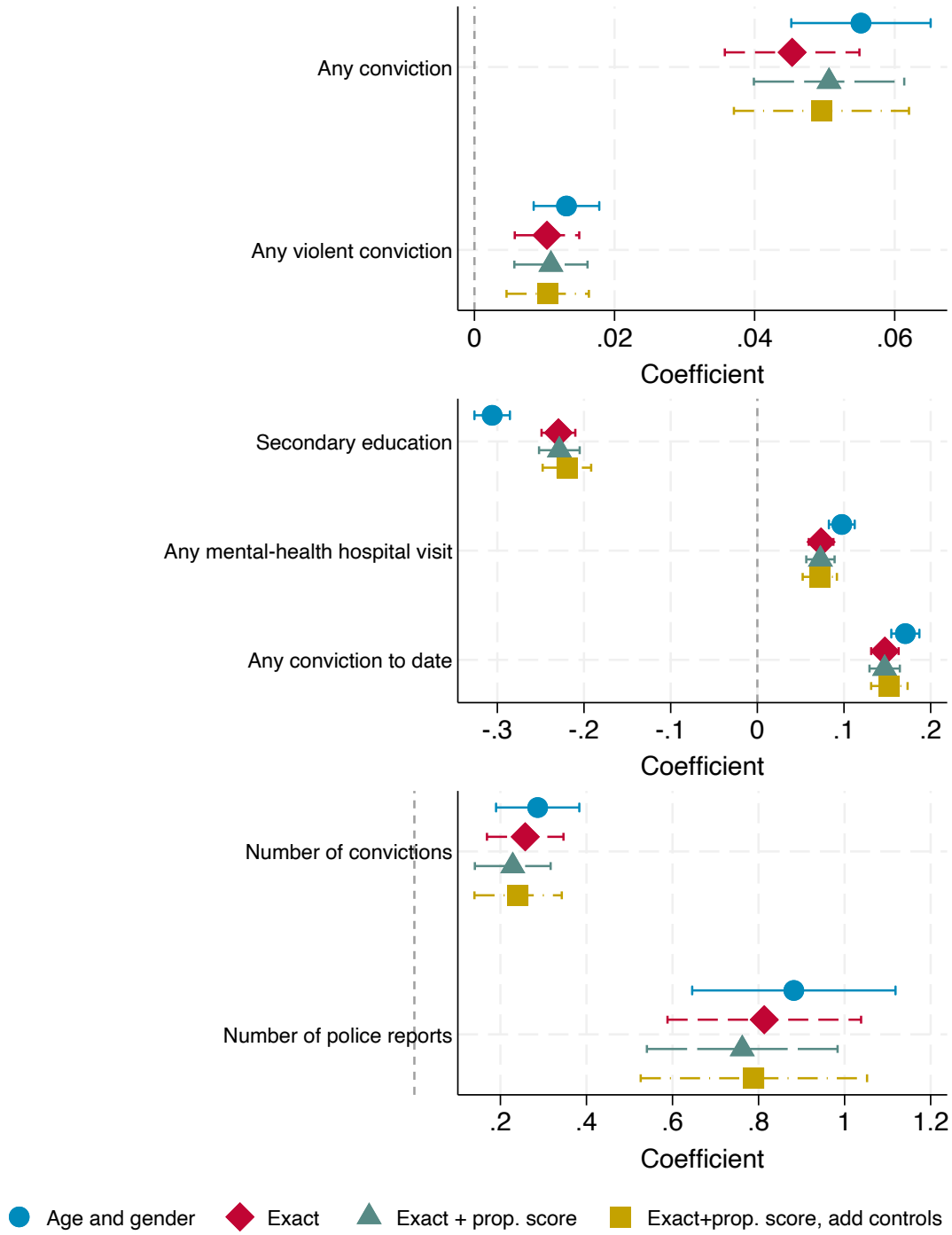


Figure A.14: Adult estimates across age-and-gender matching, exact matching, exact-plus-propensity-score matching, and exact-plus-propensity-score matching with additional pre-event controls.

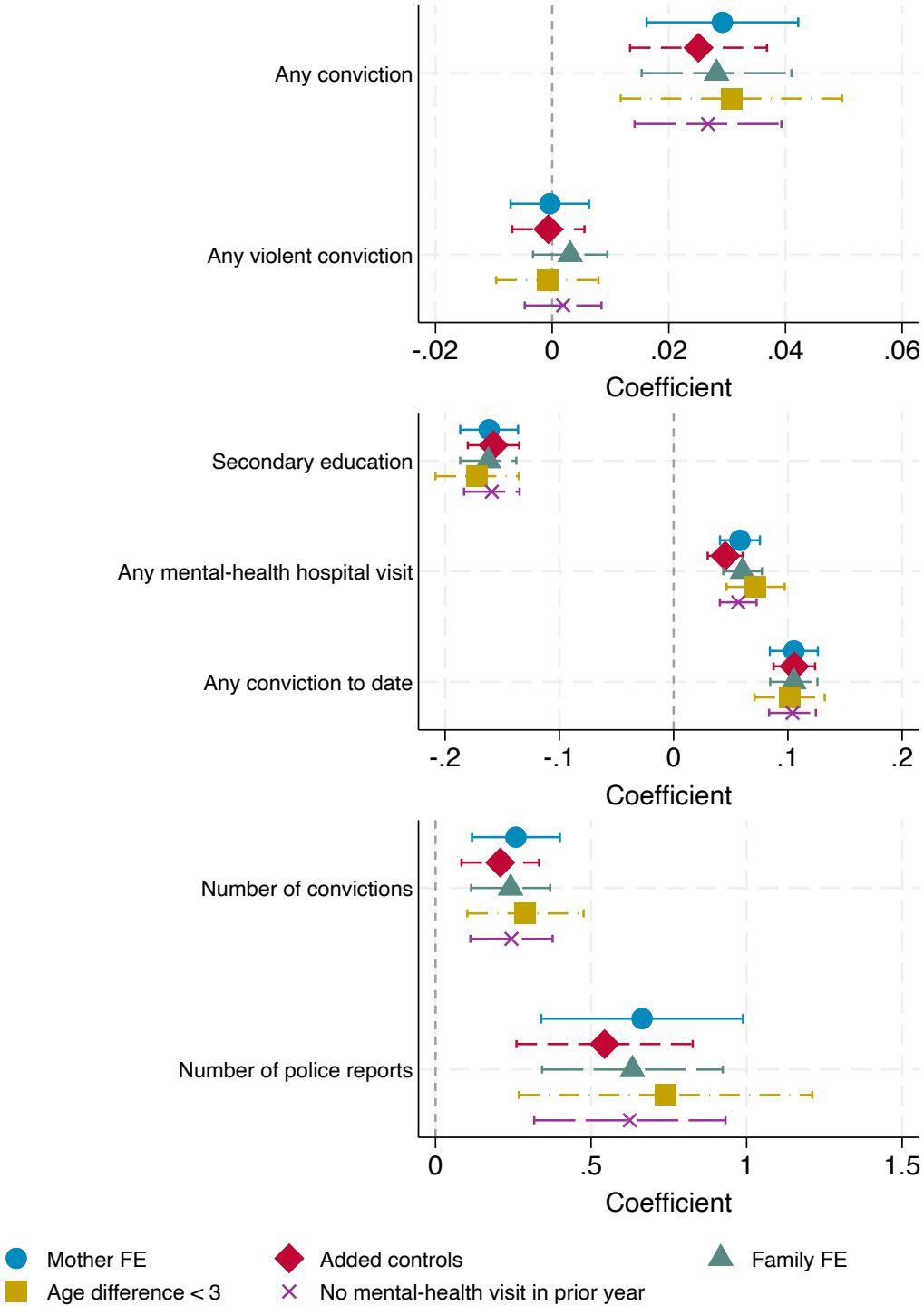
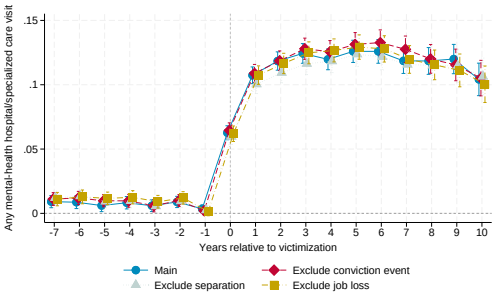
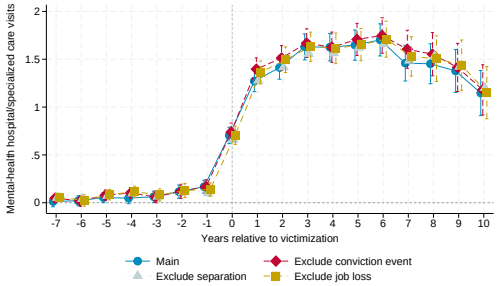


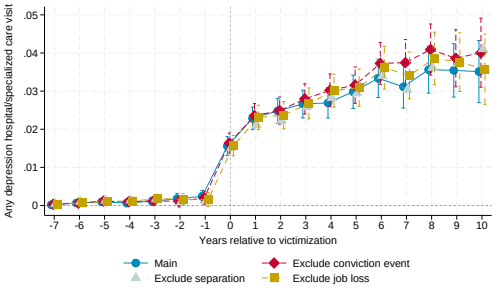
Figure A.15: Sibling estimates across alternative specifications. Points show coefficient estimates; lines show 95% confidence intervals.



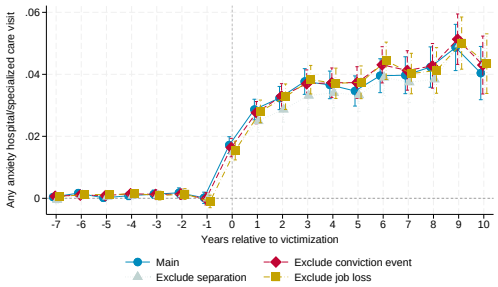
(a) Any mental health diagnosis



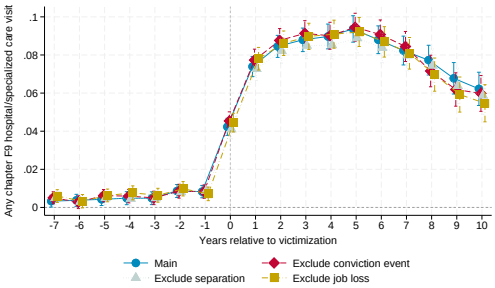
(b) N of specialist visits with MH diagnosis.



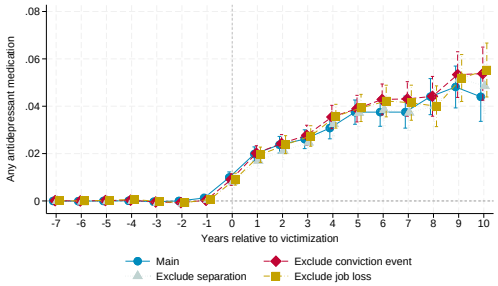
(c) Estimates: Any mood disorder diagnosis



(d) Any anxiety diagnosis



(e) Estimates: Any ADHD/conduct disorder (ICD10 F9) diagnosis



(f) Any antidepressant prescription

Figure A.16: Estimates for the main outcomes after excluding victims with simultaneous family events. See Section 4.1 for the matching specification.

	(1)	(2)	(3)	(4)	(5)
	Cum. any conv.	Cum. any conv.	Cum. any conv.	Cum. any conv.	Cum. any conv.
Victim	0.0980***	0.106***	0.105***	0.100***	0.104***
	(0.00969)	(0.00929)	(0.0105)	(0.0113)	(0.0104)
Measure age	21	21	21	21	21
Group	All	All	All	All	All
N	13239	13239	11125	9072	10034
Victim mean (SD)	.228 (.419)	.228 (.419)	.228 (.419)	.225 (.417)	.234 (.423)
Controls	Mother FE	+health controls	Fam FE	Age diff $<$ 5	No MH pre

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

Table A.7: Sibling estimates for any conviction by age 21 under alternative specifications. Column 2 adds pre-event healthcare controls, column 3 includes mother-by-father fixed effects, column 4 restricts the sibling age difference to four years or less, and column 5 restricts the sample to children with no pre-event mental-health diagnosis. Reported means and standard deviations refer to the victim sample.

A.5 Methodological details